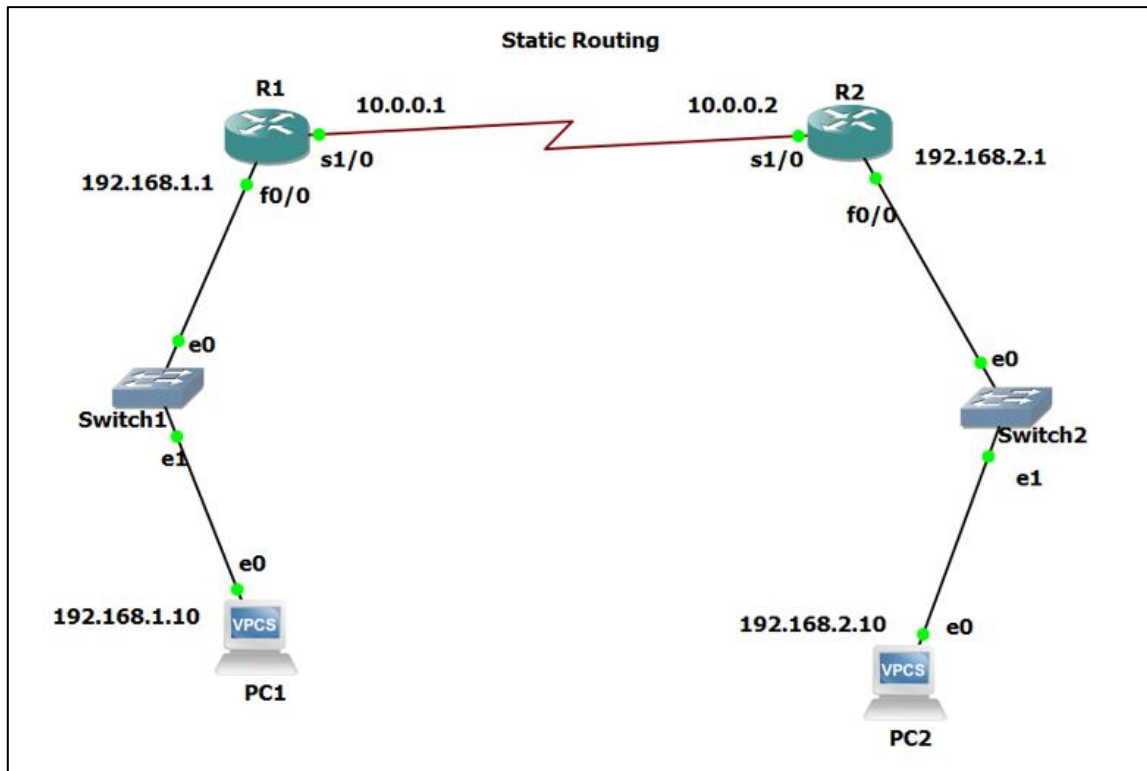


Practical 1A

Aim : Implement the concept of static routing

Topology :



Scenario: Suppose that your company has 2 branches located in Delhi (R1) and Mumbai (R2).

As the administrator of the network, you are tasked to connect them so that employees in the two LANs can communicate with each other. After careful consideration you decided to connect them via static route.

R1 >>>> Router Configuration

```
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int s1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shut
R1(config-if)#exit
```

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*Mar 1 00:07:59.367: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:08:00.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#int s1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shut
R1(config-if)#
*Mar 1 00:11:20.059: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config-if)#
*Mar 1 00:11:21.063: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config-if)#
*Mar 1 00:11:42.067: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-if)#exit
R1(config)#
```

R2 >>>> Router configuration

```
R2(config)#int f0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int s1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
```

```
R2(config)#int f0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Mar 1 00:14:20.467: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:14:21.467: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#
R2(config)#int s1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:16:03.899: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R2(config)#
*Mar 1 00:16:04.903: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
```

Static Routing on Routers

R1 >>>>

R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2

R1(config)#do sh ip route

```
R1(config)### static route to R2
R1(config)#ip route 192.168.2.0 10.0.0.2
% Incomplete command.

R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
C      10.0.0.0 is directly connected, Serial1/0
C      192.168.1.0/24 is directly connected, FastEthernet0/0
S      192.168.2.0/24 [1/0] via 10.0.0.2
R1(config)#
```

R2 >>>>

R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1

R2(config)#do sh ip route

```
R2(config)### Static route to R1
R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
C      10.0.0.0 is directly connected, Serial1/0
S      192.168.1.0/24 [1/0] via 10.0.0.1
C      192.168.2.0/24 is directly connected, FastEthernet0/0
R2(config)#
```

Manually set an IP on PC1

PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

Manually set an IP on PC2

PC2> ip 192.168.2.10 255.255.255.0 192.168.2.1

```
PC2> ip 192.168.2.10 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

Try to ping each far end network

Ping Command form PC1 towards R1, R2, PC2

```
PC1> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=15.369 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=15.750 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=15.832 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=16.168 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=15.332 ms

PC1> ### ping to R2
PC1> pint 192.168.2.1
Bad command: "pint 192.168.2.1". Use ? for help.

PC1> ping 192.168.2.1
84 bytes from 192.168.2.1 icmp_seq=1 ttl=254 time=46.496 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=254 time=46.380 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=254 time=45.399 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=254 time=45.404 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=254 time=45.664 ms

PC1> ### ping PC1 to PC2
PC1> ping 192.168.2.10
192.168.2.10 icmp_seq=1 timeout
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=61.298 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=60.515 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=60.617 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=60.945 ms
```

Ping Command form PC2 towards R1, R2, PC1

```
PC2> ## Ping R1
PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=254 time=45.502 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=254 time=45.357 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=254 time=45.795 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=254 time=46.222 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=254 time=45.330 ms

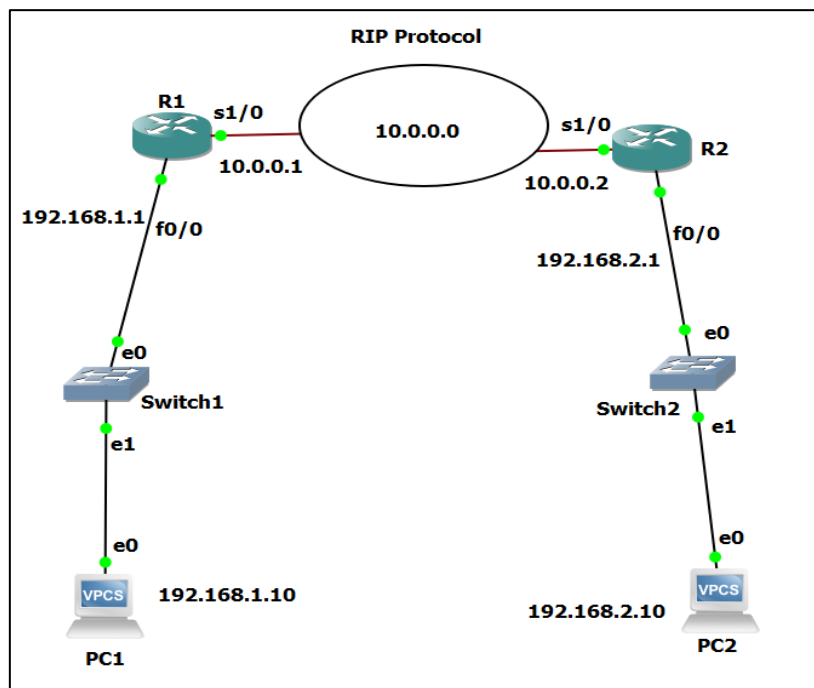
PC2> ## Ping R2
PC2> ping 192.168.2.1
84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=15.648 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=15.814 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=15.410 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=15.335 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=15.307 ms

PC2> PING PC 2 to PC 1
Bad command: "PING PC 2 to PC 1". Use ? for help.

PC2> ## PING PC 2 to PC 1
PC2> ping 192.168.1.10
192.168.1.10 icmp_seq=1 timeout
192.168.1.10 icmp_seq=2 timeout
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=61.374 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=61.800 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=62.146 ms
```

Practical 1B

Aim: Dynamic Routing Protocol (RIP)



IP and Interface Configuration

R1 >>>>

```
R1(config)#### Interface towards S1 and PC1
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#### Interface towards S1 and PC1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R1(config)### Interface towards R2
R1(config)#int s1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R1(config)### Interface towards R2
R1(config)#int s1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

R2 >>>>

```
R2#conf t
R2(config)### interface towards S2 and PC2
R2(config)#int f0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)### interface towards S2 and PC2
R2(config)#int f0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

```
R2(config)### Interface towards R1
R2(config)#int s1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R2(config)### Interface towards R1
R2(config)#int s1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

Manually set an IP on PC1

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
```

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

Manually set an IP on PC2

```
PC2> ip 192.168.2.10 255.255.255.0 192.168.2.1
```

```
PC2> ip 192.168.2.10 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

Enabling RIP Routing

```
R1(config)#router rip
R1(config-router)#network 192.168.1.0
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
```

```
R1(config)#### Enable RIP Routing
R1(config)#router rip
R1(config-router)#network 192.168.1.0
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#
*Mar 1 00:15:40.971: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
```

```
R1#show ip route
```

```
R1#write memory
Building configuration...
[OK]
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/24 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
C       192.168.1.0/24 is directly connected, FastEthernet0/0
R       192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:16, Serial1/0
R1#
```

R2(config)#### Enabling RIP Routing

```
R2(config)#router rip
R2(config-router)#network 192.168.2.0
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R2(config-router)#end
```

```
R2(config)#### Enabling RIP Routing
R2(config)#router rip
R2(config-router)#network 192.168.2.0
R2(config-router)#network 10.0.0.0
R2(config-router)#no auto-summary
R2(config-router)#end
```

```
R2#show ip route
```

```
R2#write memory
Building configuration...
[OK]
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/24 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
R       192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:18, Serial1/0
C       192.168.2.0/24 is directly connected, FastEthernet0/0
R2#
```

Ping the End Devices (PCs)

PC1> ping 192.168.2.10

```
PC1> #### Ping to PC2 from PC1
PC1> ping 192.168.2.10
192.168.2.10 icmp_seq=1 timeout
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=62.089 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=60.651 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=48.019 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=61.531 ms
```

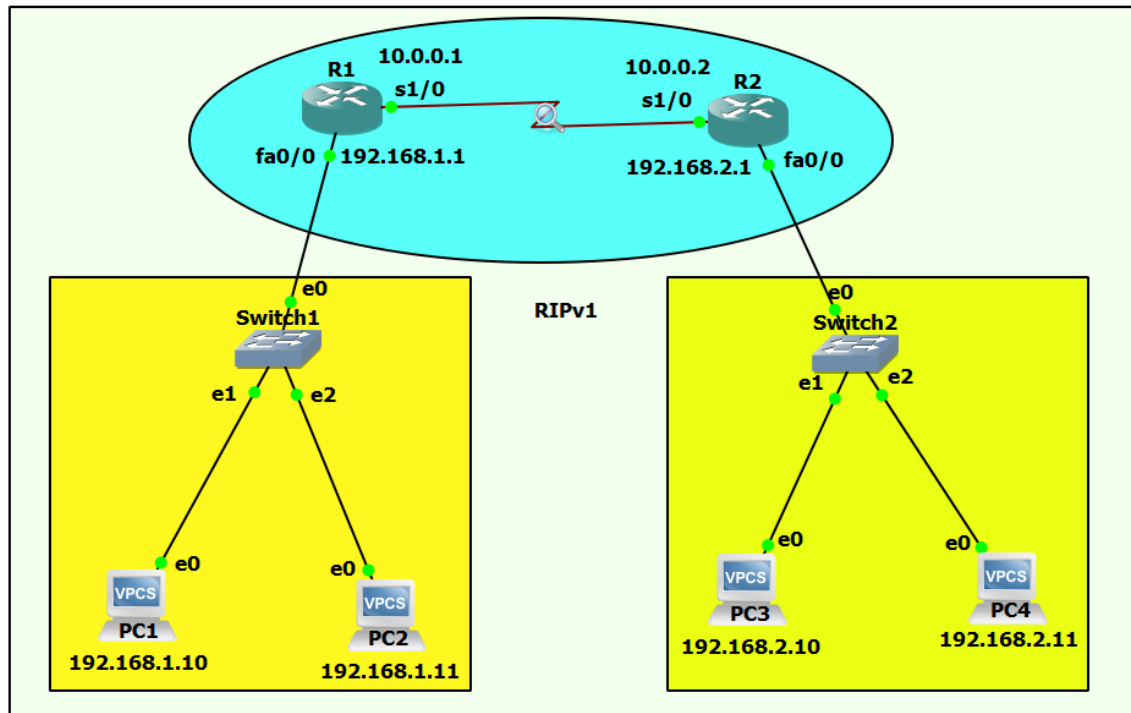
PC2> ping 192.168.1.10

```
PC2> ### Ping PC1 from PC2
PC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=62.523 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=46.207 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=46.399 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=46.735 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=62.282 ms
```

Practical 2A

Aim: Implement the concept of RIPv1 and RIPv2 routing protocol

RIPv1 Topology



R1 Config

- conf t
- int s1/0
- ip address 10.0.0.1 255.0.0.0
- clock rate 64000
- no shut
- exit
- int fa0/0
- ip address 192.168.1.1 255.255.255.0
- no shut
- exit

PC1 Config

- ip 192.168.1.10 255.255.255.0 192.168.1.1

PC2 Config

- ip 192.168.1.11 255.255.255.0 192.168.1.1

R2 Config

- conf t
- int s1/0
- ip address 10.0.0.2 255.0.0.0
- clock rate 64000
- no shut
- exit
- int fa0/0
- ip address 192.168.2.1 255.255.255.0
- no shut
- exit

PC3 Config

- ip 192.168.2.10 255.255.255.0 192.168.2.1

PC4 Config

- ip 192.168.2.11 255.255.255.0 192.168.2.1

R1 Routing

- conf t
- router rip
- network 10.0.0.0
- network 192.168.1.0
- network 192.168.2.0
- end
- do sh ip route

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:02, Serial1/0
```

R2 Routing

- conf t
- router rip
- network 10.0.0.0
- network 192.168.1.0
- network 192.168.2.0
- end
- do sh ip route

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial1/0
R    192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:09, Serial1/0
C    192.168.2.0/24 is directly connected, FastEthernet0/0
```

Result**PC1 -> PC2, PC3, PC4**

```
PC1> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.862 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.726 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.805 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.055 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.817 ms

PC1> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=61.265 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=64.700 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=48.181 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=48.625 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=46.931 ms

PC1> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=62 time=60.979 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=62 time=46.810 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=62 time=61.286 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=62 time=61.044 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=62 time=48.257 ms
```

PC2-> PC1, PC3, PC4

```
PC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=61.963 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=61.124 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=75.978 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=60.735 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=62.697 ms

PC3> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=62 time=62.248 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=62 time=60.853 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=62 time=61.526 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=62 time=60.562 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=62 time=61.692 ms

PC3> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=64 time=0.499 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=64 time=0.542 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=64 time=0.999 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=64 time=0.494 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=64 time=0.764 ms
```

PC3 -> PC1, PC2, PC4

```
PC4> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=46.253 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=60.664 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=61.722 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=62.701 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=62.694 ms

PC4> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=62 time=61.000 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=62 time=61.014 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=62 time=60.945 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=62 time=60.633 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=62 time=61.909 ms

PC4> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=64 time=0.528 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=64 time=1.130 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=64 time=0.528 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=64 time=0.737 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=64 time=1.582 ms
```

PC4 -> PC1, PC2, PC3

```
PC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=0.547 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=0.548 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=0.568 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=0.654 ms

PC2> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=60.930 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=61.967 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=61.372 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=60.520 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=61.415 ms

PC2> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=62 time=76.049 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=62 time=61.153 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=62 time=60.856 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=62 time=61.396 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=62 time=62.219 ms
```


R1 -> PC1, PC2

```
R1#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/13/24 ms
R1#ping 192.168.1.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/32 ms
```

R2 -> PC3, PC4

```
R2#ping 192.168.2.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/14/16 ms
R2#ping 192.168.2.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.11, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/19/28 ms
```

R1 -> R2

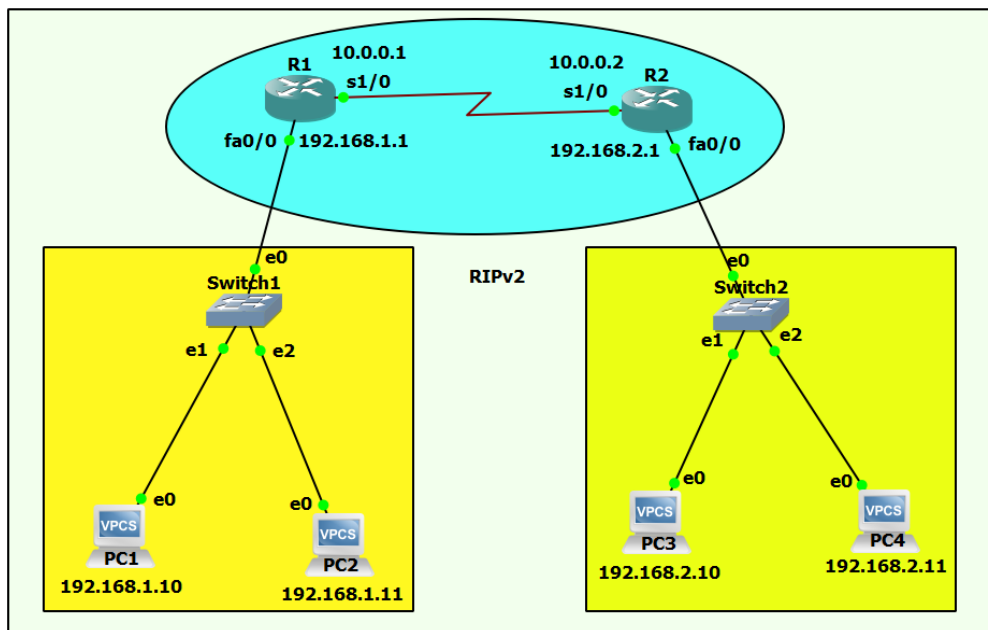
```
R1#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/60/68 ms
```

R2 -> R1

```
R2#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/41/80 ms
```

RIPv2 Topology**R1 Config**

- conf t
- int s1/0
- ip address 10.0.0.1 255.0.0.0
- clock rate 64000
- no shut
- exit
- int fa0/0
- ip address 192.168.1.1 255.255.255.0
- no shut
- exit

PC1 Config

- ip 192.168.1.10 255.255.255.0 192.168.1.1

PC2 Config

- ip 192.168.1.11 255.255.255.0 192.168.1.1

R2 Config

- conf t
- int s1/0
- ip address 10.0.0.2 255.0.0.0
- clock rate 64000
- no shut
- exit

- int fa0/0
- ip address 192.168.2.1 255.255.255.0
- no shut
- exit

PC3 Config

- ip 192.168.2.10 255.255.255.0 192.168.2.1

PC4 Config

- ip 192.168.2.11 255.255.255.0 192.168.2.1

R1 Routing v2

- conf t
- router rip
- version 2
- network 10.0.0.0
- network 192.168.1.0
- network 192.168.2.0
- end

- conf t
- do sh ip route

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:02, Serial1/0
```

R2 Routing v2

- conf t
- router rip
- version 2
- network 10.0.0.0
- network 192.168.1.0
- network 192.168.2.0
- end

- conf t
- do sh ip route

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, Serial1/0
R    192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:09, Serial1/0
C    192.168.2.0/24 is directly connected, FastEthernet0/0
```

Result**PC1 -> PC2, PC3, PC4**

```
PC1> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.862 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.726 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.805 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.055 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.817 ms

PC1> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=61.265 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=64.700 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=48.181 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=48.625 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=46.931 ms

PC1> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=62 time=60.979 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=62 time=46.810 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=62 time=61.286 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=62 time=61.044 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=62 time=48.257 ms
```

PC3 -> PC1, PC2, PC4

```
PC3> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=61.963 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=61.124 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=75.978 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=60.735 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=62.697 ms

PC3> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=62 time=62.248 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=62 time=60.853 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=62 time=61.526 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=62 time=60.562 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=62 time=61.692 ms

PC3> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=64 time=0.499 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=64 time=0.542 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=64 time=0.999 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=64 time=0.494 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=64 time=0.764 ms
```

R1 -> PC1, PC2

```
R1#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/13/24 ms
R1#ping 192.168.1.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/32 ms
```

R1 -> R2

```
R1#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/60/68 ms
```

R2 -> R1

```
R2#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/41/80 ms
```

PC2-> PC1, PC3, PC4

```
PC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=0.547 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=0.548 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=0.568 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=0.654 ms

PC2> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=60.930 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=61.967 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=61.372 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=60.520 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=61.415 ms

PC2> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=62 time=76.049 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=62 time=61.153 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=62 time=60.856 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=62 time=61.396 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=62 time=62.219 ms
```

PC4 -> PC1, PC2, PC3

```
PC4> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=46.253 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=60.664 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=61.722 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=62.701 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=62.694 ms

PC4> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=62 time=61.000 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=62 time=61.014 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=62 time=60.945 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=62 time=60.633 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=62 time=61.909 ms

PC4> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=64 time=0.528 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=64 time=1.130 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=64 time=0.528 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=64 time=0.737 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=64 time=1.582 ms
```

R2 -> PC3, PC4

```
R2#ping 192.168.2.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/14/16 ms
R2#ping 192.168.2.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.11, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/19/28 ms
```

On Router R2, execute the debug IP rip command to check the RIPv2 configuration and you will see a log as shown in the image below

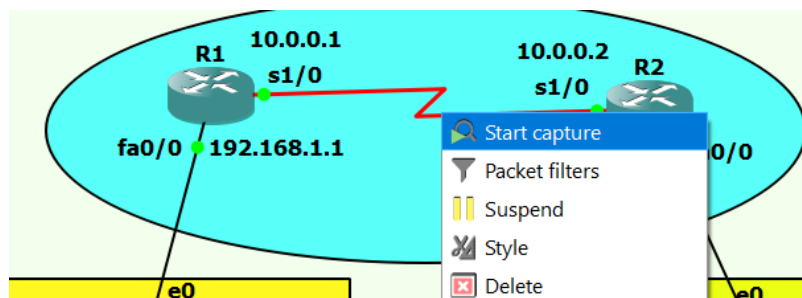
➤ do debug ip rip

```
R2(config)#do debug ip rip
RIP protocol debugging is on
R2(config)#
*Mar 1 01:15:38.267: RIP: received v2 update from 10.0.0.1 on Serial1/0
*Mar 1 01:15:38.271: 192.168.1.0/24 via 0.0.0.0 in 1 hops
R2(config)#
*Mar 1 01:15:49.411: RIP: sending v2 update to 224.0.0.9 via Serial1/0 (10.0.0.2)
*Mar 1 01:15:49.411: RIP: build update entries
*Mar 1 01:15:49.411: 192.168.2.0/24 via 0.0.0.0, metric 1, tag 0
R2(config)#
*Mar 1 01:15:52.083: RIP: sending v2 update to 224.0.0.9 via FastEthernet0/0 (192.168.2.1)
*Mar 1 01:15:52.083: RIP: build update entries
*Mar 1 01:15:52.083: 10.0.0.0/8 via 0.0.0.0, metric 1, tag 0
*Mar 1 01:15:52.087: 192.168.1.0/24 via 0.0.0.0, metric 2, tag 0
R2(config)#
*Mar 1 01:16:07.407: RIP: received v2 update from 10.0.0.1 on Serial1/0
*Mar 1 01:16:07.407: 192.168.1.0/24 via 0.0.0.0 in 1 hops
R2(config)#
*Mar 1 01:16:15.891: RIP: sending v2 update to 224.0.0.9 via Serial1/0 (10.0.0.2)
*Mar 1 01:16:15.891: RIP: build update entries
*Mar 1 01:16:15.891: 192.168.2.0/24 via 0.0.0.0, metric 1, tag 0
```

➤ do undebug ip rip

```
R2(config)#do undebug ip rip
RIP protocol debugging is off
```

We can use the Wireshark program to verify RIPv2. Start the Wireshark program I in between Serial communication line. We can view the RIPv2 (224.0.0.9) output from the image below. After clicking on Start capture the next screen opens



In this screen we can see RIPv2 output on the destination 224.0.0.9

Packet capture

Link type: Cisco HDLC

File name: R1_Serial10_to_R2_Serial10

☒ Start the capture visualization program

OK Cancel

Capturing from Standard input [R1 Serial1/0 to R2 Serial1/0]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.0.1	224.0.0.9	RIPv2	56	Response
2	2.838468	N/A	N/A	SLARP	24	Line keepal
3	2.980584	N/A	N/A	SLARP	24	Line keepal
4	12.835225	N/A	N/A	SLARP	24	Line keepal
5	12.977554	N/A	N/A	SLARP	24	Line keepal
6	14.977739	10.0.0.2	224.0.0.9	RIPv2	56	Response
7	18.520483	N/A	N/A	CDP	322	Device ID:
8	22.828688	N/A	N/A	SLARP	24	Line keepal
9	22.987395	N/A	N/A	SLARP	24	Line keepal

Frame 1: 56 bytes on wire (448 bits), 56 b...

Cisco HDLC

Internet Protocol Version 4, Src: 10.0.0.1

User Datagram Protocol, Src Port: 520, Dst...

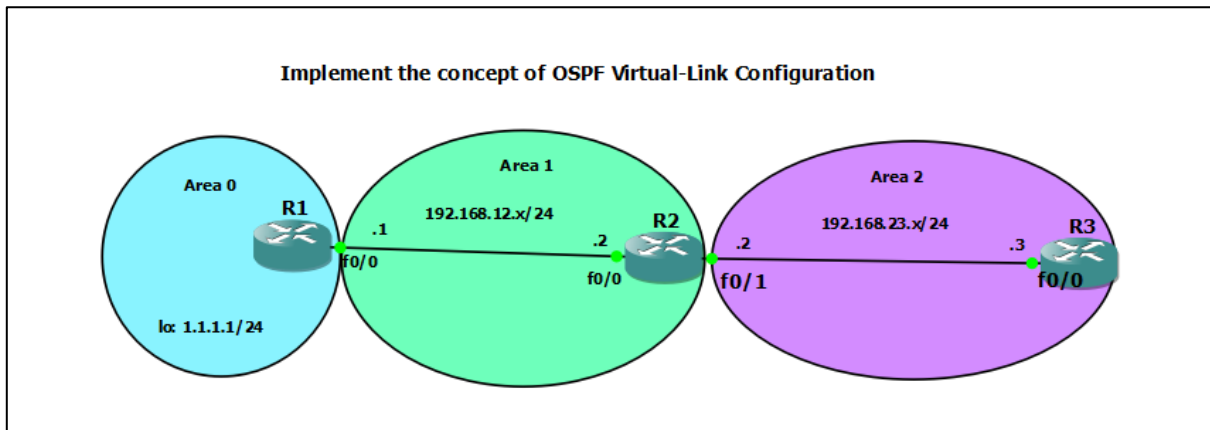
Routing Information Protocol

Standard input: <live capture in progress> Packets: 9 Profile: Default

Practical No 3A

Aim: Implement the concept of OSPF Virtual-Link Configuration

Topology



R1 >>> IP and Network configuration

```
R1#enable
R1#config t
R1(config)#ip cef
R1(config)#int lo0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#int f0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#duplex auto
R1(config-if)#speed auto
R1(config-if)#exit
R1(config)#end
```

```
R1#
R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip cef
R1(config)#int lo0
R1(config-if)#
*Mar 1 00:00:50.807: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#int f0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*Mar 1 00:02:21.475: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:22.475: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#duplex auto
R1(config-if)#speed auto
R1(config-if)#exit
R1(config)#end
```

R2 >>> IP and Network configuration

```
R2#enable
R2#conf t
R2(config)#ip cef
R2(config)#int lo0
R2(config-if)#exit
R2(config)#int f0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#duplex auto
R2(config-if)#speed auto
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 192.168.23.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#duplex auto
R2(config-if)#speed auto
R2(config-if)#exit
R2(config)#end
```

```
R2#enable
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip cef
R2(config)#int lo0
R2(config-if)#
*Mar 1 00:02:43.287: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)#exit
R2(config)#int f0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Mar 1 00:03:46.131: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:47.131: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#duplex
% Incomplete command.
R2(config-if)#auto duplex
^
% Invalid input detected at '^' marker.
R2(config-if)#duplex auto
R2(config-if)#speed auto
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 192.168.23.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Mar 1 00:10:13.371: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:10:14.371: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config-if)#duplex auto
R2(config-if)#speed auto
R2(config-if)#exit
R2(config)#end
```

R3 >>> IP and Network configuration

```

R3#enable
R3#config t
R3(config)#ip cef
R3(config)#int lo0
R3(config-if)#ip address 172.16.0.1 255.255.255.0
R3(config-if)#exit
R3(config)#int lo1
R3(config-if)#ip address 172.16.1.1 255.255.255.0
R3(config-if)#exit
R3(config)#int lo2
R3(config-if)#ip address 172.16.2.1 255.255.255.0
R3(config-if)#exit
R3(config)#int lo3
R3(config-if)#ip address 172.16.3.1 255.255.255.0
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.23.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#duplex auto
R3(config-if)#speed auto
R3(config-if)#exit
R3(config)#end

```

```

R3#
R3#enable
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip cef
R3(config)#int lo0
R3(config-if)#
*Mar 1 00:07:35.239: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 172.16.0.1 255.255.255.0
R3(config-if)#exit
R3(config)#int lo1
R3(config-if)#i
*Mar 1 00:08:49.339: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R3(config-if)#ip address 172.16.1.1 255.255.255.0
R3(config-if)#exit
R3(config)#int lo2
R3(config-if)#
*Mar 1 00:09:52.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up
R3(config-if)#ip address 172.16.2.1 255.255.255.0
R3(config-if)#exit
R3(config)#int lo3
R3(config-if)#
*Mar 1 00:11:01.611: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback3, changed state to up
R3(config-if)#ip address 172.16.3.1 255.255.255.0
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.23.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#
*Mar 1 00:13:18.563: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:13:19.563: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to up
R3(config-if)#duplex auto
R3(config-if)#speed auto
R3(config-if)#exit
R3(config)#end
R3#

```

SHOW IP ROUTE

R1#show ip route

```

R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
C    1.0.0.0/24 is subnetted, 1 subnets
C      1.1.1.0 is directly connected, Loopback0

```

R2#show ip route

```

R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
C    192.168.23.0/24 is directly connected, FastEthernet0/1

```

R3#show ip route

```

R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

      172.16.0.0/24 is subnetted, 4 subnets
C      172.16.0.0 is directly connected, Loopback0
C      172.16.1.0 is directly connected, Loopback1
C      172.16.2.0 is directly connected, Loopback2
C      172.16.3.0 is directly connected, Loopback3
C    192.168.23.0/24 is directly connected, FastEthernet0/0

```


OSPF Configuration >>>

```
R1(config)#router ospf 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.12.0 0.0.0.255 area 1
R1(config-router)#exit
R1(config)#end
```

```
R1(config)#router ospf 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.12.0 0.0.0.255 area 1
R1(config-router)#exit
R1(config)#end
R1#
*Mar 1 00:24:52.503: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Mar 1 00:26:04.547: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 1
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 1
*Mar 1 00:24:04.495: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
```

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.23.0 0.0.0.255 area 2
R3(config-router)#exit
R3(config)#
```

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.23.0 0.0.0.255 area 2
R3(config-router)#exit
R3(config)#
*Mar 1 00:20:42.683: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config)#exit
```

Showing IP OSPF Neighbor

```
R1#show ip ospf neighbor
```

```
R1#show ip ospf neighbor

Neighbor ID    Pri   State       Dead Time   Address        Interface
192.168.23.2    1     FULL/BDR    00:00:33    192.168.12.2   FastEthernet0/0
R1#
```

```
R2#show ip ospf neighbor
```

```
R2#show ip ospf neighbor

Neighbor ID    Pri   State       Dead Time   Address        Interface
1.1.1.1         1     FULL/DR     00:00:38    192.168.12.1   FastEthernet0/0
172.16.3.1      1     FULL/BDR    00:00:35    192.168.23.3   FastEthernet0/1
R2#
```

```
R3#show ip ospf neighbor
```

```
R3#show ip ospf neighbor

Neighbor ID    Pri   State       Dead Time   Address        Interface
192.168.23.2    1     FULL/DR     00:00:39    192.168.23.2   FastEthernet0/0
R3#
```

Virtual Linking

```
R1#config t
R1(config)#router ospf 1
R1(config-router)#area 1 virtual-link 192.168.23.2
```

```
R1### Virtual Linking
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#area 1 virtual-link 192.168.23.2
R1(config-router)#
*Mar 1 00:35:17.475: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on OSPF_VL0 from LOADING to FULL, Loading Done
R1(config-router)#
```

```
R2#config t
R2(config)#router ospf 1
R2(config-router)#area 1 virtual-link 1.1.1.1
```

```
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#area 1 virtual-link 1.1.1.1
R2(config-router)#
*Mar 1 00:33:17.419: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on OSPF_VL0 from LOADING to FULL, Loading Done
R2(config-router)#
```


R1#ping 192.168.23.3

```
R1#ping 192.168.23.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 48/55/60 ms
R1#ping 192.168.23.0
```

R1#show ip ospf virtual-links

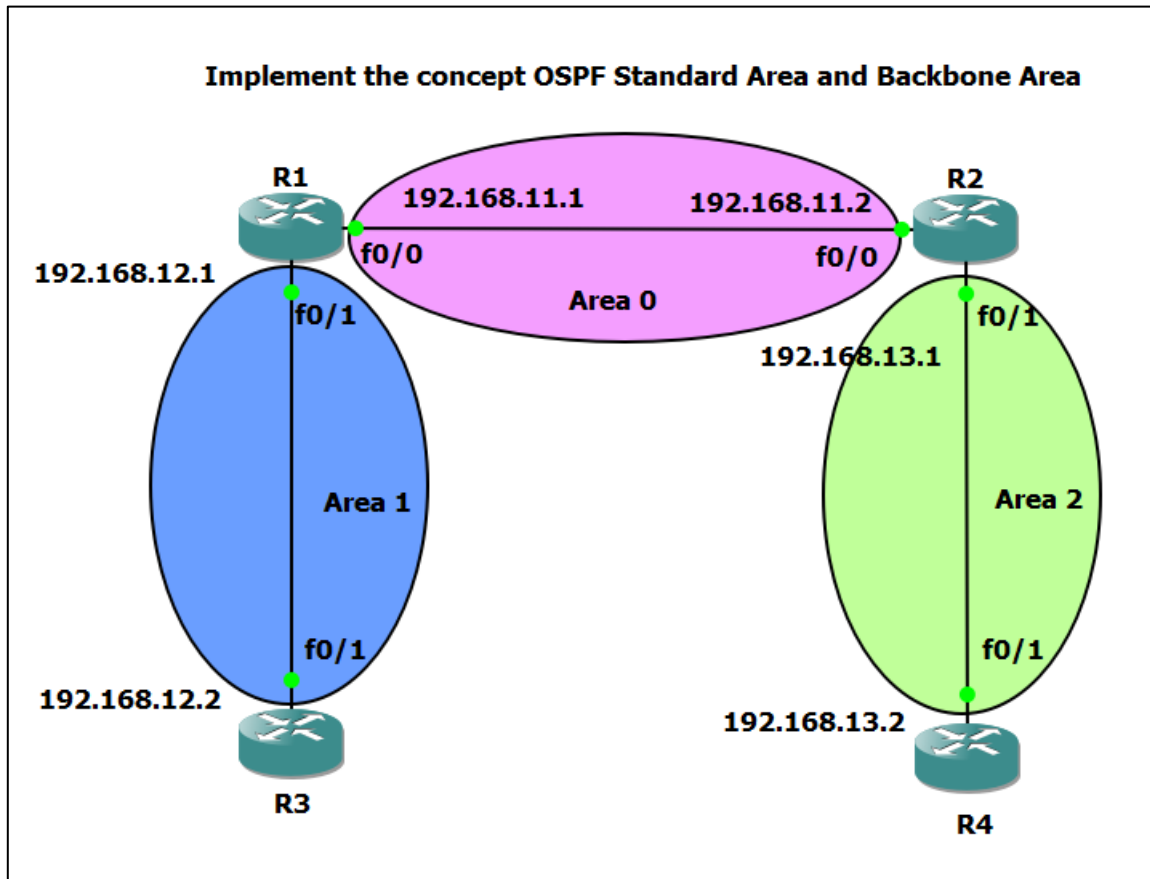
```
R1#show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 192.168.23.2 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface FastEthernet0/0, Cost of using 10
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04
  Adjacency State FULL (Hello suppressed)
  Index 1/2, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
R1#
```

R2#show ip ospf virtual-links

```
R2#show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 1.1.1.1 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface FastEthernet0/0, Cost of using 10
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Adjacency State FULL (Hello suppressed)
  Index 1/3, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
R2#
```

Practical 3 B

Aim: Implement the concept OSPF Standard Area and Backbone Area



IP configuration of the routers R1, R2, R3, R4 as follows:

R1 >>>> IP and Network Configuration

```
R1#enable
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.11.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int f0/1
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R1#
R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.11.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:02:05.175: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:06.183: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#int f0/1
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ex
*Mar 1 00:03:07.199: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:08.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config-if)#exit
```

R2 >>>> IP and Network Configuration

```
R2#enable
R2#config t
R2(config)#int f0/0
R2(config-if)#ip address 192.168.11.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 192.168.13.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R2#enable
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.11.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:09.627: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:10.627: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#int f0/1
R2(config-if)#ip address 192.168.13.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:05:03.111: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:05:04.111: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

R3 >>>> IP and Network Configuration

```
R3#enable
R3#config t
R3(config)#int loopback 0
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int f0/1
R3(config-if)#ip address 192.168.12.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R3#enable
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int loopback 0
R3(config-if)#ip
*Mar 1 00:04:36.475: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
R3(config)#int f0/1
R3(config-if)#ip address 192.168.12.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
*Mar 1 00:07:24.427: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:07:25.427: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

R4 >>>> IP and Network Configuration

```
R4#enable
R4#config t
R4(config)#int loopback 0
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int f0/1
R4(config-if)#ip address 192.168.13.2 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
```

```
R4#enable
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int loopback 0
R4(config-if)#
*Mar 1 00:06:24.519: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
R4(config)#int f0/1
R4(config-if)#ip address 192.168.13.2 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
*Mar 1 00:08:11.983: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:08:12.983: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Configure the ospf for every router as follows:

R1 and R2 for the **backbone area**:

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.11.0 0.0.0.255 area 0
R1(config-router)#exit
```

```
R1(config)#### OSPF configuration
R1(config)#router ospf 1
R1(config-router)#network 192.168.11.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#
*Mar 1 00:19:11.931: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.11.0 0.0.0.255 area 0
R2(config-router)#exit
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.11.0 0.0.0.255 area 0
R2(config-router)#
*Mar 1 00:17:21.803: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.12.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
R2(config-router)#exit
```

configure R1 and R3 for **area 1**:

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 1
R1(config-router)#exit
```

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 1
R1(config-router)#exit
R1(config)#
R1(config)#
*Mar 1 00:27:08.263: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on FastEthernet0/1 from LOADING to FULL, Loading Done
```

```
R3(config)#router ospf 1
R3(config-router)#network 3.3.3.3 0.0.0.255 area 1
R3(config-router)#network 192.168.12.0 0.0.0.255 area 1
R3(config-router)#exit
```

```
R3(config)#router ospf 1
R3(config-router)#network 3.3.3.3 0.0.0.255 area 1
R3(config-router)#network 192.168.12.0 0.0.0.255 area 1
R3(config-router)#exit
R3(config-router)#exit
*Mar 1 00:22:58.143: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.12.1 on FastEthernet0/1 from LOADING to FULL, Loading Done
```

Configure R2 and R4 for area 2:

```
R2(config)#router ospf 1
```

```
R2(config-router)#network 192.168.13.0 0.0.0.255 area 2
```

```
R2(config-router)#exit
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.13.0 0.0.0.255 area 2
R2(config-router)#exit
R2(config)#
*Mar 1 00:29:29.299: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet0/1 from LOADING to FULL, Loading Done
```

```
R4(config)#router ospf 1
```

```
R4(config-router)#network 192.168.13.0 0.0.0.255 area 2
```

```
R4(config-router)#network 4.4.4.4 0.0.0.0 area 2
```

```
R4(config-router)#exit
```

```
R4(config)#router ospf 1
R4(config-router)#network 192.168.13.0 0.0.0.255 area 2
R4(config-router)#network
*Mar 1 00:24:49.023: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from LOADING to FULL, Loading Done
R4(config-router)#network 4.4.4.4 0.0.0.0 area 2
R4(config-router)#exit
```

Verification of the network**Check the OSPF neighbors**

```
R1(config)#do show ip ospf neighbor
```

R1 has formed a neighbor adjacency with R2 and R3.

```
R1(config)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.13.1	1	FULL/BDR	00:00:32	192.168.11.2	FastEthernet0/0
3.3.3.3	1	FULL/BDR	00:00:31	192.168.12.2	FastEthernet0/1

```
R1(config)#
```

```
R2(config)#do show ip ospf neighbor
```

```
R2(config)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.12.1	1	FULL/DR	00:00:32	192.168.11.1	FastEthernet0/0
4.4.4.4	1	FULL/BDR	00:00:37	192.168.13.2	FastEthernet0/1

R2 has formed neighbor adjacencies with R1 and R4.

The show ip ospf neighbor command, however, doesn't tell anything about the areas that are used, So we check it using the show ip ospf neighbor detail command.

```
R2(config)#do show ip ospf neighbor detail
```

```
R2(config)#do show ip ospf neighbor detail
Neighbor 192.168.12.1, interface address 192.168.11.1
In the area 0 via interface FastEthernet0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.11.1 BDR is 192.168.11.2
Options is 0x52
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 00:18:11
Index 1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 4.4.4.4, interface address 192.168.13.2
In the area 2 via interface FastEthernet0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.13.1 BDR is 192.168.13.2
Options is 0x52
LLS Options is 0x1 (LR)
Dead timer due in 00:00:33
Neighbor is up for 00:06:04
Index 1/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
R2(config)#
```

Above you can see that interface FastEthernet0/0 is in area 0 and FastEthernet0/1 is in area 2.

Another good command to find area information is show ip protocols:

```
R2(config)#do show ip protocols
```

```
R2(config)#do show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.13.1
  It is an area border router
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.11.0 0.0.0.255 area 0
    192.168.13.0 0.0.0.255 area 2
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.4.4.4          110          00:07:09
    192.168.13.1     110          00:09:40
    192.168.12.1     110          00:09:45
  Distance: (default is 110)
```

Above you can see which networks belong to which area:

Network 192.168.11.0 in area 0.

Network 192.168.13.0 in area 2.

Check the routing tables:

R1(config)#do show ip route ospf

```
R1(config)#do show ip route ospf
O IA 192.168.13.0/24 [110/20] via 192.168.11.2, 00:14:06, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O    3.3.3.3 [110/11] via 192.168.12.2, 00:16:37, FastEthernet0/1
    4.0.0.0/32 is subnetted, 1 subnets
O IA    4.4.4.4 [110/21] via 192.168.11.2, 00:11:35, FastEthernet0/0
R1(config)#
```

Above we see three OSPF entries. The first one is for 3.3.3.3/32, the loopback interface of R3. R1 has also learned about 4.4.4.4/32 and 192.168.24.0/24. These two entries show up as O IA since they are inter-area routes.

R2(config)#do show ip route ospf

```
R2(config)#do show ip route ospf
O IA 192.168.12.0/24 [110/20] via 192.168.11.1, 00:11:03, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O IA    3.3.3.3 [110/21] via 192.168.11.1, 00:11:03, FastEthernet0/0
    4.0.0.0/32 is subnetted, 1 subnets
O    4.4.4.4 [110/11] via 192.168.13.2, 00:08:27, FastEthernet0/1
R2(config)#
```

Above we see that R2 has learned about 3.3.3.3/32 and 192.168.13.0/24 which are inter-area routes. 4.4.4.4/32 is an intra-area route.

R3(config)#do show ip route ospf

```
R3(config)#do show ip route ospf
O IA 192.168.13.0/24 [110/30] via 192.168.12.1, 00:15:58, FastEthernet0/1
    4.0.0.0/32 is subnetted, 1 subnets
O IA    4.4.4.4 [110/31] via 192.168.12.1, 00:13:27, FastEthernet0/1
O IA 192.168.11.0/24 [110/20] via 192.168.12.1, 00:18:29, FastEthernet0/1
R3(config)#
```

Everything that R3 has learned is from another area, that's why we only see inter-area routes here.

R4(config)#do show ip route ospf

```
R4(config)#do show ip route ospf
O IA 192.168.12.0/24 [110/30] via 192.168.13.1, 00:14:43, FastEthernet0/1
    3.0.0.0/32 is subnetted, 1 subnets
O IA    3.3.3.3 [110/31] via 192.168.13.1, 00:14:43, FastEthernet0/1
O IA 192.168.11.0/24 [110/20] via 192.168.13.1, 00:14:43, FastEthernet0/1
R4(config)#
```

Just to be sure, let's try a quick ping between R3 and R4 to prove that our multi-area OSPF configuration is working:

R3#do ping 4.4.4.4 source 3.3.3.3

```
R3(config)#do ping 4.4.4.4 source 3.3.3.3

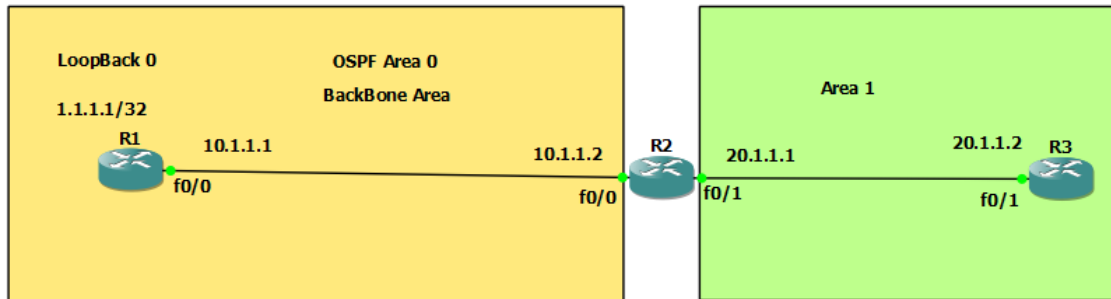
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
Packet sent with a source address of 3.3.3.3
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/87/96 ms
R3(config)#
```

Practical 3C

Aim: Implement the concept OSPF Stub and Totally Stubby Area

Topology:

The concept OSPF Stub and Totally Stubby Area



Configure the IPv4 address on the physical interface on the routers:

R1 >>>> IP and Network Configuration

```
R1#enable
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
*Mar 1 00:02:24.163: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:25.163: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

R2 >>>> IP and Network Configuration

```
R2#enable
R2#config t
R2(config)#int f0/0
R2(config-if)#ip address 10.1.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#int f0/1
R2(config-if)#ip address 20.1.1.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R2#enable
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 10.1.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#int f0/1
R2(config-if)#ip address 20.1.1.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
*Mar 1 00:03:11.143: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:12.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:44.563: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:45.563: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config-if)#exit
```

R3 >>>> IP and Network Configuration

```
R3#enable
R3#config t
R3(config)#int f0/1
R3(config-if)#ip address 20.1.1.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R3#enable
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip address 20.1.1.2 255.255.255.0
^
% Invalid input detected at '^' marker.
R3(config)#int f0/1
R3(config-if)#ip address 20.1.1.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
*Mar 1 00:04:43.847: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:04:44.847: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config-if)#exit
```

Configure a loopback interface on R1 with an IPv4 address:

```
R1(config)#int loopback 0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
```

```
R1(config)#int loopback 0
R1(config-if)#ip ad
*Mar 1 00:07:35.663: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
```


Configuring the OSPF process on all three routers with the command router OSPF <process-id>:

R1(config)#router ospf 1
R1(config-router)#exit

```
R1(config)#router ospf 1
R1(config-router)#exit
```

R2(config)#router ospf 1
R2(config-router)#exit

```
R2(config)#router ospf 1
R2(config-router)#exit
```

R3(config)#router ospf 1
R3(config-router)#exit

```
R3(config)#router ospf 1
R3(config-router)#exit
```

Enabling OSPF on the interfaces with the interface level sub-command IP OSPF <process-id>**area <area-id> :**

R1(config)#int loopback 0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int f0/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#no shut

```
R1(config)#int loopback 0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int f0/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#
*Mar 1 00:15:38.619: %OSPF-5-ADJCHG: Process 1, Nbr 20.1.1.1 on FastEthernet0/0 from LOADING to FULL,
Loading Done
```

R2(config)#int f0/0
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut

```
R2(config-if)#int f0/0
R2(config-if)#ip ospf 1 area 1
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:15:57.963: %OSPF-5-ADJCHG: Process 1, Nbr 20.1.1.2 on FastEthernet0/1 from LOADING to FULL,
Loading Done
```

R3(config)#int f0/1
R3(config-if)#ip ospf 1 area 1
R3(config-if)#no shut

```
R3(config)#int f0/1
R3(config-if)#ip ospf 1 area 1
R3(config-if)#no shut
R3(config-if)#ex
*Mar 1 00:15:48.019: %OSPF-5-ADJCHG: Process 1, Nbr 20.1.1.1 on FastEthernet0/1 from LOADING to FULL,
Loading Done
```

Configuring area 2 as a stub area with the router sub-command area <area-id> stub

R2(config)#router ospf 1
R2(config-router)#area 1 stub
R2(config-router)#exit

```
R2(config)#router ospf 1
R2(config-router)#area 1 stub
R2(config-router)#
*Mar 1 00:18:31.567: %OSPF-5-ADJCHG: Process 1, Nbr 20.1.1.2 on FastEthernet0/1 from FULL to DOWN, Nei
ghbor Down: Adjacency forced to reset
R2(config-router)#exit
```

R3(config-router)#area 1 stub
R3(config-router)#exit

```
R3(config)#router ospf 1
R3(config-router)#
*Mar 1 00:17:57.879: %OSPF-5-ADJCHG: Process 1, Nbr 20.1.1.1 on FastEthernet0/1 from FULL to DOWN, Nei
ghbor Down: Dead timer expired
R3(config-router)#area 1 stub
R3(config-router)#exit
*Mar 1 00:18:18.019: %OSPF-5-ADJCHG: Process 1, Nbr 20.1.1.1 on FastEthernet0/1 from LOADING to FULL,
Loading Done
```

Verifying OSPF configuration on the routers:

R1(config)#do show ip protocol

```
R1(config)#do show ip protocol
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    Routing on Interfaces Configured Explicitly (Area 0):
      FastEthernet0/0
      Loopback0
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110           00:08:49
    20.1.1.1         110           00:06:40
  Distance: (default is 110)
```

R2(config)#do show ip protocol

```
R2(config)#do show ip protocol
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 20.1.1.1
  It is an area border router
  Number of areas in this router is 2. 1 normal 1 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    FastEthernet0/0
  Routing on Interfaces Configured Explicitly (Area 1):
    FastEthernet0/1
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance         Last Update
    1.1.1.1           110              00:04:22
    20.1.1.1          110              00:07:10
  Distance: (default is 110)
```

R3(config)#do show ip protocol

```
R3(config)#do show ip protocol
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 20.1.1.2
  Number of areas in this router is 1. 0 normal 1 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 1):
    FastEthernet0/1
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance         Last Update
    20.1.1.1          110              00:03:56
    20.1.1.2          110              00:06:28
  Distance: (default is 110)
```

Checking the IPv4 routing table on the routers and seeing OSPF route entry:

R1(config)#do show ip route

```
R1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback0
  20.0.0.0/24 is subnetted, 1 subnets
O IA    20.1.1.0 [110/20] via 10.1.1.2, 00:08:35, FastEthernet0/0
  10.0.0.0/24 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, FastEthernet0/0
```

R2(config)#do show ip route

```
R2(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/11] via 10.1.1.1, 00:06:02, FastEthernet0/0
  20.0.0.0/24 is subnetted, 1 subnets
C       20.1.1.0 is directly connected, FastEthernet0/1
  10.0.0.0/24 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, FastEthernet0/0
```

R3(config)#do show ip route

```
R3(config)#do show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
```

```
ia - IS-IS inter area, * - candidate default, U - per-user static route
```

```
o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is 20.1.1.1 to network 0.0.0.0
```

```
1.0.0.0/32 is subnetted, 1 subnets
```

```
O IA 1.1.1.1 [110/21] via 20.1.1.1, 00:05:19, FastEthernet0/1
```

```
20.0.0.0/24 is subnetted, 1 subnets
```

```
C 20.1.1.0 is directly connected, FastEthernet0/1
```

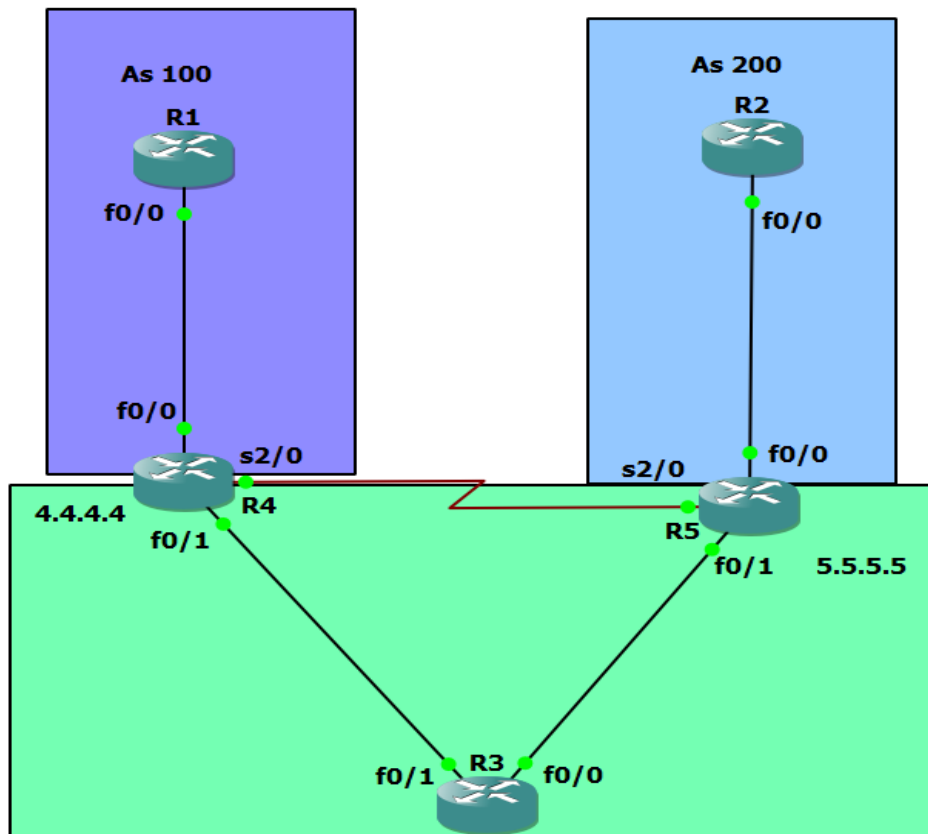
```
10.0.0.0/24 is subnetted, 1 subnets
```

```
O IA 10.1.1.0 [110/20] via 20.1.1.1, 00:05:19, FastEthernet0/1
```

```
O*IA 0.0.0.0/0 [110/11] via 20.1.1.1, 00:05:19, FastEthernet0/1
```

Practical 4A

Aim: - Implement the concept of BGP routing protocol



Configure the IPv4 address on the physical interface on the routers:

R1 >>>> IP and Network Configuration

```
R1#enable
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.14.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int loopback 1
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#end
```

```
R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.14.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*Mar 1 00:03:10.931: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:11.931: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#int loopback 1
R1(config-if)#
*Mar 1 00:03:55.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#end
R1#
*Mar 1 00:41:28.639: %SYS-5-CONFIG_I: Configured from console by console
```

R2 >>>> IP and Network Configuration

```
R2#enable
R2#config t
R2(config)#int f0/0
R2(config-if)#ip address 192.168.25.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int loopback 2
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
```

```
R2#enable
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.25.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Mar 1 00:04:31.795: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:04:32.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#int loopback 2
R2(config-if)#ip
*Mar 1 00:04:55.335: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#end
R2#
*Mar 1 00:40:19.435: %SYS-5-CONFIG_I: Configured from console by console
```

R3 >>>> IP and Network Configuration

```
R3#config t
R3(config)#int f0/1
R3(config-if)#ip address 192.168.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.35.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int loopback 0
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#no shut
R3(config-if)#exit
```

```
R3#enable
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int f0/1
R3(config-if)#ip address 192.168.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#
*Mar 1 00:06:08.723: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:06:09.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.35.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
*Mar 1 00:07:57.031: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:07:58.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#int loopback 0
R3(config-if)#
*Mar 1 00:08:26.399: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#no shut
R3(config-if)#exit
```

```
R3(config)#router ospf 1
R3(config-router)#network 3.3.3.3 0.0.0.0 area 0
R3(config-router)#network 192.168.34.0 0.0.0.255 area 0
R3(config-router)#network 192.168.35.0 0.0.0.255 area 0
R3(config-router)#exit
```

```
R3(config)#router ospf 1
R3(config-router)#network 3.3.3.3 0.0.0.0 area 0
R3(config-router)#network 192.168.34.0 0.0.0.255 area 0
R3(config-router)#network 192.168.35.0 0.0.0.255 area 0
R3(config-router)#no shut
^
% Invalid input detected at '^' marker.
R3(config-router)#no shutdown
^
% Invalid input detected at '^' marker.
R3(config-router)#exit
R3(config)#
R3(config)#
*Mar 1 00:28:35.879: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet0/1 from LOADING to FULL, Loading Done
R3(config)#
*Mar 1 00:39:55.067: %OSPF-5-ADJCHG: Process 1, Nbr 5.5.5.5 on FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config)#end
R3#
*Mar 1 00:41:37.511: %SYS-5-CONFIG_I: Configured from console by console
```

R4 >>>> IP and Network Configuration

```
R4#enable
R4#config t
R4(config)#int f0/0
R4(config-if)#ip address 192.168.14.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int s2/0
R4(config-if)#ip address 192.168.45.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int f0/1
R4(config-if)#ip address 192.168.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int loopback 0
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
```

```
R4#enable
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)#ip address 192.168.14.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#
*Mar 1 00:10:40.283: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:10:41.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config-if)#exit
R4(config)#int s2/0
R4(config-if)#ip address 192.168.45.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config-if)#exit
*Mar 1 00:11:56.199: %LINK-3-UPDOWN: Interface Serial2/0, changed state to up
R4(config-if)#exit
*Mar 1 00:11:57.203: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R4(config-if)#exit
R4(config)#int f0/1
R4(config-if)#
*Mar 1 00:12:22.183: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down
R4(config-if)#ip address 192.168.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#
*Mar 1 00:13:00.067: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:13:01.067: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4(config-if)#exit
R4(config)#int loopback 0
R4(config-if)#
*Mar 1 00:13:28.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
```

```
R4(config)#router ospf 1
R4(config-router)#network 4.4.4.4 0.0.0.0 area 0
R4(config-router)#network 192.168.34.0 0.0.0.255 area 0
R4(config-router)#router ospf 1
R4(config-router)#network 192.168.45.0 0.0.0.255 area 0
R4(config-router)#exit
R4(config)#end
```

```
R4(config)#router ospf 1
R4(config-router)#network 4.4.4.4 0.0.0.0 area 0
R4(config-router)#network 192.168.34.0 0.0.0.255 area 0
R4(config-router)#router ospf 1
*Mar 1 00:23:35.351: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on FastEthernet0/1 from LOADING to FULL, Loading Done
R4(config-router)#network 192.168.45.0 0.0.0.255 area 0
R4(config-router)#no shut
^
% Invalid input detected at '^' marker.
R4(config-router)#exit
R4(config)#
R4(config)#
*Mar 1 00:29:22.155: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R4(config)#end
R4#
*Mar 1 00:36:29.779: %SYS-5-CONFIG_I: Configured from console by console
```

R5 >>>> IP and Network Configuration

```

R5#enable
R5#config t
R5(config)#int f0/0
R5(config-if)#ip address 192.168.25.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#int f0/1
R5(config-if)#ip address 192.168.35.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#int s2/0
R5(config-if)#ip address 192.168.15.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#exit
R5(config)#int loopback 0
R5(config-if)#ip address 5.5.5.5 255.255.255.255
R5(config-if)#no shut
R5(config-if)#exit

```

```

R5(config)#router ospf 1
R5(config-router)#network 5.5.5.5 0.0.0.0 area 0

R5(config-router)#network 192.168.35.0 0.0.0.255 area 0

R5(config-router)#network 192.168.45.0 0.0.0.255 area 0

R5(config-router)#exit

R5(config)#end

```

```

R5#enable
R5#config t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#int f0/0
R5(config-if)#ip address 192.168.25.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#
*Mar 1 00:18:38.283: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:18:39.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R5(config-if)#exit
R5(config)#int f0/1
R5(config-if)#ip address 192.168.35.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#
*Mar 1 00:19:40.667: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:19:41.667: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R5(config-if)#exit
R5(config)#int s2/0
R5(config-if)#ip address 192.168.15.5 255.255.255.0
R5(config-if)#no shut
R5(config-if)#
*Mar 1 00:20:27.375: %LINK-3-UPDOWN: Interface Serial2/0, changed state to up
R5(config-if)#
*Mar 1 00:20:28.379: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R5(config-if)#exit
R5(config)#
R5(config)#int loopback 0
R5(config-if)#
*Mar 1 00:21:04.455: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R5(config-if)#ip address 5.5.5.5 255.255.255.255
R5(config-if)#no shut
R5(config-if)#exit

```

```

R5(config)#router ospf 1
R5(config-router)#network 5.5.5.5 0.0.0.0 area 0
R5(config-router)#network 192.168.35.0 0.0.0.255 area 0
R5(config-router)#network
*Mar 1 00:25:54.239: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on FastEthernet0/1 from LOADING to FULL, Loading Done
R5(config-router)#network 192.168.45.0 0.0.0.255 area 0
R5(config-router)#no shut
^
% Invalid input detected at '^' marker.

R5(config-router)#no shut
^
% Invalid input detected at '^' marker.

R5(config-router)#exit
R5(config)#end
R5#
*Mar 1 00:27:17.143: %SYS-5-CONFIG_I: Configured from console by console

```

R1#### Show IP Routes

R1#show ip route

```

R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback1
C    192.168.14.0/24 is directly connected, FastEthernet0/0

```

R2#show ip route

```

R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    2.0.0.0/32 is subnetted, 1 subnets
C       2.2.2.2 is directly connected, Loopback2
C    192.168.25.0/24 is directly connected, FastEthernet0/0

```


R3#show ip route

```
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3 is directly connected, Loopback0
O       192.168.45.0/24 [110/74] via 192.168.34.4, 00:10:51, FastEthernet0/1
    4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/11] via 192.168.34.4, 00:16:38, FastEthernet0/1
    5.0.0.0/32 is subnetted, 1 subnets
O       5.5.5.5 [110/11] via 192.168.35.5, 00:05:19, FastEthernet0/0
C       192.168.34.0/24 is directly connected, FastEthernet0/1
C       192.168.35.0/24 is directly connected, FastEthernet0/0
```

R4#show ip route

```
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C       192.168.14.0/24 is directly connected, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/11] via 192.168.34.3, 00:19:12, FastEthernet0/1
C       192.168.45.0/24 is directly connected, Serial2/0
    4.0.0.0/32 is subnetted, 1 subnets
C       4.4.4.4 is directly connected, Loopback0
    5.0.0.0/32 is subnetted, 1 subnets
O       5.5.5.5 [110/21] via 192.168.34.3, 00:07:54, FastEthernet0/1
C       192.168.34.0/24 is directly connected, FastEthernet0/1
O       192.168.35.0/24 [110/20] via 192.168.34.3, 00:19:14, FastEthernet0/1
```

R5#show ip route

```
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/11] via 192.168.35.3, 00:07:58, FastEthernet0/1
C       192.168.15.0/24 is directly connected, Serial2/0
O       192.168.45.0/24 [110/84] via 192.168.35.3, 00:07:58, FastEthernet0/1
C       192.168.25.0/24 is directly connected, FastEthernet0/0
    4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/21] via 192.168.35.3, 00:07:58, FastEthernet0/1
    5.0.0.0/32 is subnetted, 1 subnets
C       5.5.5.5 is directly connected, Loopback0
O       192.168.34.0/24 [110/20] via 192.168.35.3, 00:08:00, FastEthernet0/1
O       192.168.35.0/24 is directly connected, FastEthernet0/1
```

EBGP Configuration

R1#config t

R1(config)#router bgp 100

R1(config-router)#network 1.1.1.0 mask 255.255.255.255

R1(config-router)#neighbor 192.168.14.4 remote-as 300

R1(config-router)#end

```
R1##### EBGp Configuration
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 100
R1(config-router)#network 1.1.1.0 mask 255.255.255.255
R1(config-router)#neighbor 192.168.14.4 remote-as 300
R1(config-router)#end
R1#
*Mar  1 00:50:40.019: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Mar  1 00:54:44.111: %BGP-5-ADJCHANGE: neighbor 192.168.14.4 Up
```

R2#config t

R2(config)#router bgp 200

R2(config-router)#network 2.2.2.0 mask 255.255.255.255

R2(config-router)#neighbor 192.168.25.5 remote-as 300

R2(config-router)#end

```
R2### EBGp Config
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router bgp 200
R2(config-router)#network 2.2.2.0 mask 255.255.255.255
R2(config-router)#neighbor 192.168.25.5 remote-as 300
R2(config-router)#end
R2#
*Mar  1 00:51:42.487: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Mar  1 00:58:34.775: %BGP-5-ADJCHANGE: neighbor 192.168.25.5 Up
```

R4#config t

R4(config)#router bgp 300

R4(config-router)#neighbor 192.168.14.1 remote-as 100

R4(config-router)#end

```
R4### EBGp Config
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router bgp 300
R4(config-router)#neighbor 192.168.14.1 remote-as 100
R4(config-router)#end
R4#
*Mar  1 00:51:01.679: %SYS-5-CONFIG_I: Configured from console by console
R4#
*Mar  1 00:51:03.659: %BGP-5-ADJCHANGE: neighbor 192.168.14.1 Up
```

R5##### EBGIP Config

R5#config t

R5(config)#router bgp 300

R5(config-router)#neighbor 192.168.25.2 remote-as 200

R5(config-router)#exit

R5(config)#end

```

R5##### EBGIP Config
R5#config t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router bgp 300
R5(config-router)#neighbor 192.168.25.2 remote-as 200
R5(config-router)#exit
R5(config)#end
R5#
*Mar  1 00:46:57.627: %SYS-5-CONFIG_I: Configured from console by console
R5#
*Mar  1 00:47:04.023: %BGP-5-ADJCHANGE: neighbor 192.168.25.2 Up
R5#

```

IBGP Configuration

R3#config t

R3(config)#router bgp 300

R3(config-router)#neighbor 4.4.4.4 remote-as 300

R3(config-router)#neighbor 4.4.4.4 update-source loopback 0

R3(config-router)#neighbor 5.5.5.5 remote-as 300

R3(config-router)#neighbor 5.5.5.5 update-source loopback 0

R3(config-router)#end

```

R3#### IBGP Configuration
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)#neighbor 4.4.4.4 remote-as 300
R3(config-router)#neighbor 4.4.4.4 update-source loopback 0
R3(config-router)#neighbor 5.5.5.5 remote-as 300
R3(config-router)#neighbor 5.5.5.5 update-source loopback 0
R3(config-router)#end
R3#
*Mar  1 01:05:25.951: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Mar  1 01:07:04.551: %BGP-5-ADJCHANGE: neighbor 4.4.4.4 Up
R3#
*Mar  1 01:10:44.743: %BGP-5-ADJCHANGE: neighbor 5.5.5.5 Up
R3#

```

R4#config t

R4(config)#router bgp 300

R4(config-router)#neighbor 3.3.3.3 remote-as 300

R4(config-router)#neighbor 3.3.3.3 update-source loopback 0

R4(config-router)#neighbor 5.5.5.5 remote-as 300

R4(config-router)#neighbor 5.5.5.5 update-source loopback 0

R4(config-router)#end

```

R4#### IBGP Config
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router bgp 300
R4(config-router)#neighbor 3.3.3.3 remote-as 300
R4(config-router)#neighbor 3.3.3.3
*Mar  1 01:02:04.015: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Up
R4(config-router)#neighbor 3.3.3.3 update-source loopback 0
R4(config-router)#neighbor 5.5.5.5 remote-as 300
R4(config-router)#neighbor 5.5.5.5 update-source loopback 0
R4(config-router)#end
R4#
*Mar  1 01:03:29.471: %SYS-5-CONFIG_I: Configured from console by console
R4#
*Mar  1 01:05:02.999: %BGP-5-ADJCHANGE: neighbor 5.5.5.5 Up
R4#

```

R5#config t

R5(config)#router bgp 300

R5(config-router)#neighbor 4.4.4.4 remote-as 300

R5(config-router)#neighbor 4.4.4.4 update-source loopback 0

R5(config-router)#neighbor 3.3.3.3 remote-as 300

R5(config-router)#neighbor 3.3.3.3 update-source loopback 0

R5(config-router)#end

```

R5#### IBGP Config
R5#config t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#router bgp 300
R5(config-router)#neighbor 4.4.4.4 remote-as 300
R5(config-router)#neighbor 4.4.4.4 update-source loopback 0
R5(config-router)#
*Mar  1 00:56:02.679: %BGP-5-ADJCHANGE: neighbor 4.4.4.4 Up
R5(config-router)#neighbor 3.3.3.3 remote-as 300
R5(config-router)#neighbor 3.3.3.3 update-source loopback 0
R5(config-router)#end
R5#
*Mar  1 00:56:40.271: %SYS-5-CONFIG_I: Configured from console by console
R5#
*Mar  1 00:56:43.871: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Up
R5#

```

Show BGP Neighbor

R1#show bgp neighbor

```

R1#show bgp neighbor
BGP neighbor is 192.168.14.4, remote AS 300, external link
BGP version 4, remote router ID 4.4.4.4
BGP state = Established, up for 00:15:18
Last read 00:00:18, last write 00:00:18, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(old & new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

      Sent       Rcvd
Opens:          1         1
Notifications:  0         0
Updates:         0         0
Keepalives:     18        18
Route Refresh:  0         0
Total:          19        19
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 1, neighbor version 1/0
Output queue size: 0
Index 1, Offset 0, Mask 0x2
1 update-group member

```

R2#show bgp neighbor

```
R2#show bgp neighbor
BGP neighbor is 192.168.25.5, remote AS 300, external link
BGP version 4, remote router ID 5.5.5.5
BGP state = Established, up for 00:11:50
Last read 00:00:50, last write 00:00:50, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(old & new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

              Sent      Rcvd
Opens:         1         1
Notifications: 0         0
Updates:       0         0
Keepalives:    14        14
Route Refresh: 0         0
Total:         15        15
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 1, neighbor version 1/0
Output queue size: 0
```

R3#show bgp neighbor

```
R3#show bgp neighbor
BGP neighbor is 4.4.4.4, remote AS 300, internal link
BGP version 4, remote router ID 4.4.4.4
BGP state = Established, up for 00:06:05
Last read 00:00:05, last write 00:00:05, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(old & new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

              Sent      Rcvd
Opens:         1         1
Notifications: 0         0
Updates:       0         0
Keepalives:    8         9
Route Refresh: 0         0
Total:         9         10
Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
BGP table version 1, neighbor version 1/0
Output queue size: 0
```

R4#show bgp neighbor

```
R4#show bgp neighbor
BGP neighbor is 3.3.3.3, remote AS 300, internal link
BGP version 4, remote router ID 3.3.3.3
BGP state = Established, up for 00:06:17
Last read 00:00:17, last write 00:00:17, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(old & new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

              Sent      Rcvd
Opens:         1         1
Notifications: 0         0
Updates:       0         0
Keepalives:    9         8
Route Refresh: 0         0
Total:        10         9
Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
BGP table version 1, neighbor version 1/0
Output queue size: 0
```

R5#show bgp neighbor

```
R5#show bgp neighbor
BGP neighbor is 3.3.3.3, remote AS 300, internal link
BGP version 4, remote router ID 3.3.3.3
BGP state = Established, up for 00:02:47
Last read 00:00:47, last write 00:00:47, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(old & new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

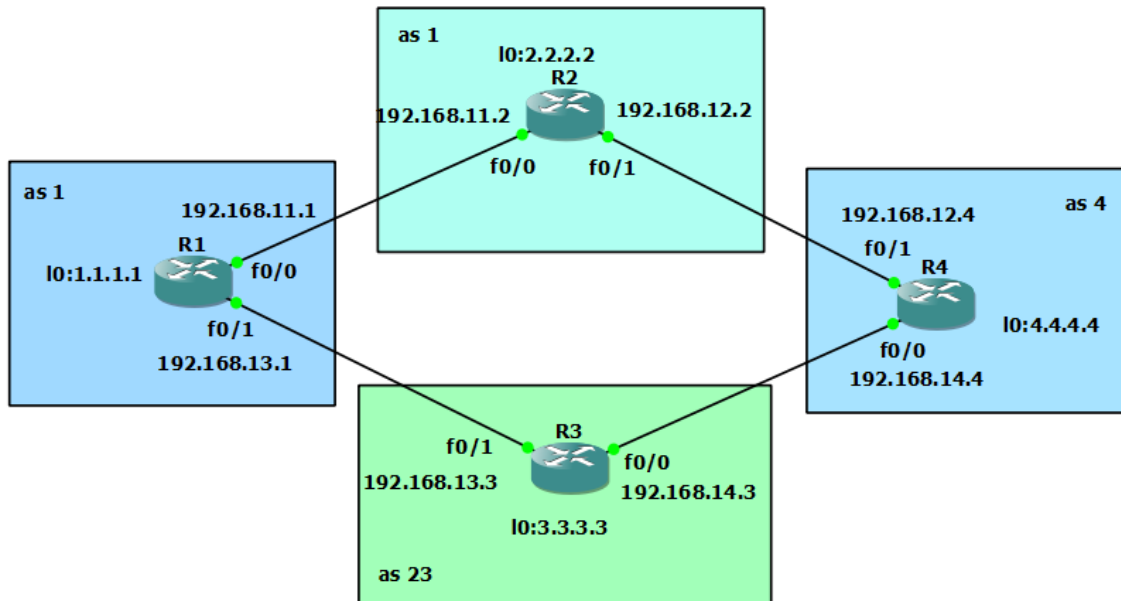
              Sent      Rcvd
Opens:         1         1
Notifications: 0         0
Updates:       0         0
Keepalives:    5         5
Route Refresh: 0         0
Total:         6         6
Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
BGP table version 1, neighbor version 1/0
Output queue size: 0
--More--
```

Practical No :4B

Aim: Implement the concept of BGP Path Attributes – MED

The concept of BGP Path Attributes – MED



IP and Interface configuration for R1, R2, R3, R4 as follows

R1 >>>> IP and Network Configuration

```

R1#config t
R1(config)#int loopback 0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int f0/0
R1(config-if)#ip address 192.168.11.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int f0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
  
```

```

R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int loopback 0
R1(config-if)#
*Mar 1 00:05:15.791: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
R1(config)#int f0/0
R1(config-if)#ip address 192.168.11.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:06:23.443: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:24.443: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#int f0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:07:13.167: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:07:14.167: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
  
```

R2 >>>> IP and Network Configuration

```

R2#config t
R2(config)#int loopback 0
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/0
R2(config-if)#ip address 192.168.11.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
  
```

```

R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int loopback 0
R2(config-if)#
*Mar 1 00:04:18.503: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
R2(config)#int f0/0
R2(config-if)#ip address 192.168.11.2 255.255.255.0
^
% Invalid input detected at '^' marker.
R2(config-if)#ip address 192.168.11.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Mar 1 00:06:26.807: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:27.807: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:08:56.363: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:08:57.363: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
  
```

R3 >>>> IP and Network Configuration

```
R3#config t
R3(config)#int loopback 0
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
R3(config)#int f0/1
R3(config-if)#ip address 192.168.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.14.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int loopback 0
R3(config-if)#
*Mar 1 00:06:49.215: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
R3(config)#int f0/1
R3(config-if)#ip address 192.168.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
*Mar 1 00:08:46.891: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:08:47.891: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.14.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
*Mar 1 00:10:42.699: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:10:43.699: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

R4 >>>> IP and Network Configuration

```
R4#config t
R4(config)#int loopback 0
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
R4(config)#int f0/1
R4(config-if)#ip address 192.168.12.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int f0/1
R4(config-if)#exit
R4(config)#int f0/0
R4(config-if)#ip address 192.168.14.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
```

```
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int loopback 0
R4(config-if)#ip a
*Mar 1 00:09:15.067: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
R4(config)#int f0/1
R4(config-if)#ip address 192.168.12.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
*Mar 1 00:11:07.587: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:11:08.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4(config)#int f0/1
R4(config-if)#exit
R4(config)#int f0/0
R4(config-if)#ip address 192.168.14.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
*Mar 1 00:12:22.043: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:12:23.043: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

R1(config)#### IP interface brief

```
R1(config)#do show ip int brief
```

```
R1(config)#### IP Interface Brief
R1(config)#do show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.11.1    YES manual up          up
Serial0/0       unassigned      YES unset  administratively down down
FastEthernet0/1 192.168.13.1    YES manual up          up
Serial0/1       unassigned      YES unset  administratively down down
Serial1/0       unassigned      YES unset  administratively down down
Serial1/1       unassigned      YES unset  administratively down down
Serial1/2       unassigned      YES unset  administratively down down
Serial1/3       unassigned      YES unset  administratively down down
Serial2/0       unassigned      YES unset  administratively down down
Serial2/1       unassigned      YES unset  administratively down down
Serial2/2       unassigned      YES unset  administratively down down
Serial2/3       unassigned      YES unset  administratively down down
```

```
R2(config)#do show ip int brief
```

```
R2(config)### IP interface brief
R2(config)#do show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.11.2    YES manual up          up
Serial0/0       unassigned      YES unset  administratively down down
FastEthernet0/1 192.168.12.2    YES manual up          up
```


R3(config)#do show ip int brief

```
R3(config)### IP interface brief
R3(config)#do show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.14.3    YES manual up          up
Serial0/0       unassigned      YES unset  administratively down down
FastEthernet0/1 192.168.13.3    YES manual up          up
```

R4(config)#do show ip int brief

```
R4(config)### IP interface brief
R4(config)#do show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.14.4    YES manual up          up
Serial0/0       unassigned      YES unset  administratively down down
FastEthernet0/1 192.168.12.4    YES manual up          up
```

Configure BGP in all the routers

R1(config)#router bgp 1

R1(config-router)#neighbor 192.168.11.2 remote-as 1

R1(config-router)#neighbor 192.168.13.3 remote-as 23

R1(config-router)#redistribute connected

R1(config-router)#exit

```
R1(config)### BGP Config
R1(config)#router bgp 1
R1(config-router)#neighbor 192.168.11.2 remote-as 1
R1(config-router)#neighbor 192.168.13.3 remote-as 23
R1(config-router)#redistribute connected
R1(config-router)#exit
R1(config)#
*Mar 1 00:44:57.763: %BGP-5-ADJCHANGE: neighbor 192.168.11.2 Up
R1(config)#
*Mar 1 00:49:50.127: %BGP-5-ADJCHANGE: neighbor 192.168.13.3 Up
```

R2(config)#router bgp 1

R2(config-router)#neighbor 192.168.11.1 remote-as 1

R2(config-router)#neighbor 192.168.12.4 remote-as 4

R2(config-router)#redistribute connected

R2(config-router)#exit

```
R2(config)### BGP Config
R2(config)#router bgp 1
R2(config-router)#neighbor 192.168.11.1 remote-as 1
R2(config-router)#neighbor 192.168.11.1 remote-as 1
*Mar 1 00:41:17.723: %BGP-5-ADJCHANGE: neighbor 192.168.11.1 Up
R2(config-router)#neighbor 192.168.12.4 remote-as 4
R2(config-router)#redistribute connected
R2(config-router)#exit
R2(config)#
*Mar 1 00:48:16.995: %BGP-5-ADJCHANGE: neighbor 192.168.12.4 Up
R2(config)#
R2#
*Mar 1 01:20:28.203: %SYS-5-CONFIG_I: Configured from console by console
```

R3(config)#router bgp 23

R3(config-router)#neighbor 192.168.13.1 remote-as 1

R3(config-router)#neighbor 192.168.14.4 remote-as 4

R3(config-router)#redistribute connected

R3(config-router)#exit

```
R3(config)### BGP Config
R3(config)#router bgp 23
R3(config-router)#neighbor 192.168.13.1 remote-as 1
R3(config-router)#neighbor 192.168.13.1 remote-as 1
*Mar 1 00:42:19.923: %BGP-5-ADJCHANGE: neighbor 192.168.13.1 Up
R3(config-router)#neighbor 192.168.14.4 remote-as 4
R3(config-router)#redistribute connected
R3(config-router)#exit
R3(config)#
*Mar 1 00:45:22.207: %BGP-5-ADJCHANGE: neighbor 192.168.14.4 Up
```

R4(config)#router bgp 4

R4(config-router)#neighbor 192.168.12.2 remote-as 1

R4(config-router)#neighbor 192.168.14.3 remote-as 23

R4(config-router)#redistribute connected

R4(config-router)#exit

```
R4(config)### BGP Config
R4(config)#router bgp 4
R4(config-router)#neighbor 192.168.12.2 remote-as 1
R4(config-router)#neighbor 192.168.12.2 remote-as 1
*Mar 1 00:42:26.579: %BGP-5-ADJCHANGE: neighbor 192.168.12.2 Up
R4(config-router)#neighbor 192.168.14.3 remote-as 23
R4(config-router)#redistribute connected
R4(config-router)#exit
R4(config)#
*Mar 1 00:43:21.931: %BGP-5-ADJCHANGE: neighbor 192.168.14.3 Up
R4(config)#
R4#
*Mar 1 01:15:44.243: %SYS-5-CONFIG_I: Configured from console by console
```

From the below R1 BGP table, we can see that the R1 prefers the path through R3 (Next Hop IP as 192.168.13.3) to reach the Network 4.4.4.4/32. Symbol ">" in-front of the network indicates the best path (router prefers to use this path to send the traffic)

R1(config)#### Show BGP Table**R1(config)#do show ip bgp**

```

R1(config)#do show ip bgp
BGP table version is 9, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*> 1.1.1.1/32      0.0.0.0          0           32768 ?
*>i 2.2.2.2/32      192.168.11.2     0          100      0 ?
*> 3.3.3.3/32      192.168.13.3     0           0 23 ?
* 4.4.4.4/32      192.168.13.3     0           0 23 4 ?
*>i 192.168.12.4    192.168.12.4     0          100      0 4 ?
* i 192.168.11.0   192.168.11.2     0          100      0 ?
*> 0.0.0.0          0.0.0.0          0          32768 ?
*>i 192.168.12.0    192.168.11.2     0          100      0 ?
* 192.168.13.0     192.168.13.3     0           0 23 ?
*> 0.0.0.0          0.0.0.0          0          32768 ?
* i 192.168.14.0   192.168.12.4     0          100      0 4 ?
*> 192.168.13.3     192.168.13.3     0           0 23 ?

```

Now, our task to change the path through R2 to reach the Network 4.4.4.4/32

Create route-map MED-TEST

Here, We have created a route-map named as MED-TEST. Next, We have used the command "match ip address 1" to match the access-list 1, and then if it matches we have set the Metric (MED) value as 500. We have created a second permit statement for the same route-map AS-PATH-TEST.

R3(config)#### Creating Route-Map

```

R3(config)#route-map MED-TEST permit
R3(config-route-map)#match ip address 1
R3(config-route-map)#set metric 500
R3(config-route-map)#exit
R3(config)#route-map MED-TEST permit 20
R3(config-route-map)#exit
R3(config)#do sh route-map

```

```

R3(config)#### Creating Route-Map
R3(config)#route-map MED-TEST permit
R3(config-route-map)#match ip address 1
R3(config-route-map)#set metric 500
R3(config-route-map)#exit
R3(config)#route-map MED-TEST permit 20
R3(config-route-map)#exit
R3(config)#do sh route-map
route-map MED-TEST, permit, sequence 10
  Match clauses:
    ip address (access-lists): 1
  Set clauses:
    metric 500
  Policy routing matches: 0 packets, 0 bytes
route-map MED-TEST, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes

```

Next create a access-list as below:

```

R3(config)#### Creating access-list
R3(config)#access-list 1 permit host 4.4.4.4
R3(config)#exit
R3#show access-lists

```

```

R3(config)#### Creating access-list
R3(config)#access-list 1 permit host 4.4.4.4
R3(config)#exit
R3#
*Mar  1 01:01:18.735: %SYS-5-CONFIG_I: Configured from console by console
R3#show access-lists
Standard IP access list 1
  10 permit 4.4.4.4

```

Applying the created route-map to bgp configuration.

Here, we have mapped the route-map MED-TEST as "OUT" to the neighbor R1 (192.168.13.1). so that it will advertise the configured Metric value (value: 500) its neighbor.

R3#### MED-TEST as out

```

R3#config t
R3(config)#router bgp 23
R3(config-router)#neighbor 192.168.13.1 route-map MED-TEST out
R3(config-router)#exit

```

```

R3#### MED-TEST as out
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 23
R3(config-router)#neighbor 192.168.13.1 route-map MED-TEST out
R3(config-router)#exit

```

You need to clear the BGP session to take the newly applied configuration to take effect. In-order to clear the bgp session, use the command "clear ip bgp * soft in"

R1(config)### Clear BGP Session: clear ip bgp * soft in

R1(config)#do clear ip bgp * soft in

R1(config)#do show ip bgp

```
R1(config)### Clear BGP Session Command : clear ip bgp * soft in
R1(config)#do clear ip bgp * soft in
R1(config)#do show ip bgp
BGP table version is 9, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

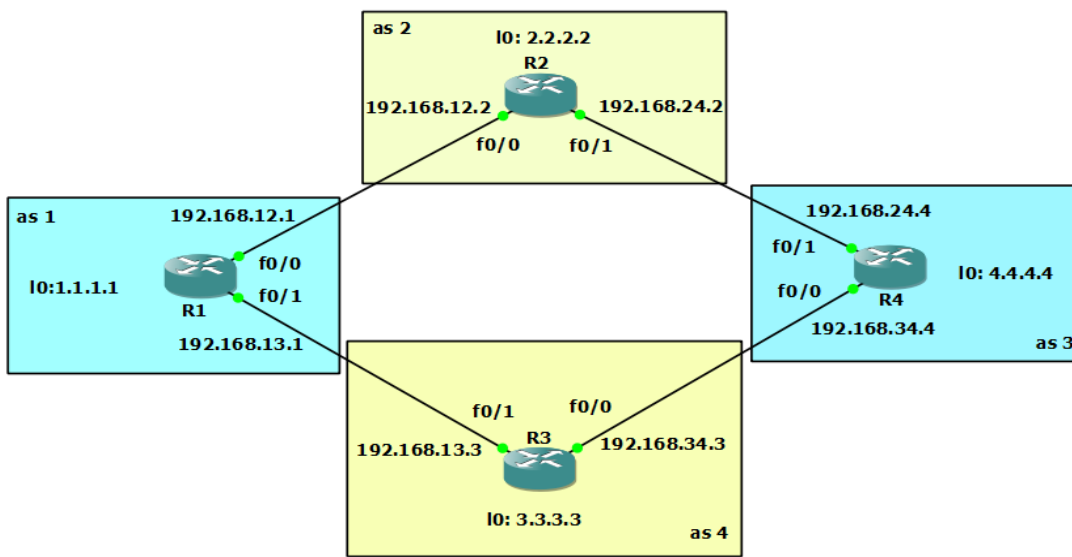
   Network          Next Hop          Metric LocPrf Weight Path
*> 1.1.1.1/32        0.0.0.0                0           32768 ?
*>i 2.2.2.2/32        192.168.11.2            0         100      0 ?
*> 3.3.3.3/32        192.168.13.3            0           0 23 ?
* 4.4.4.4/32        192.168.13.3           500           0 23 4 ?
*>i 192.168.11.0      192.168.12.4            0         100      0 4 ?
* i 192.168.11.0      192.168.11.2            0         100      0 ?
*> 0.0.0.0            0.0.0.0                0           32768 ?
*>i 192.168.12.0      192.168.11.2            0         100      0 ?
* 192.168.13.0        192.168.13.3            0           0 23 ?
*> 0.0.0.0            0.0.0.0                0           32768 ?
* i 192.168.14.0      192.168.12.4            0         100      0 4 ?
*> 192.168.13.3        192.168.13.3            0           0 23 ?
R1(config)#
```

Here, We can see that, The metric for the neighbor 192.168.13.3 (R3 router) has now changed to 500. Best path changed to R2 (Next hop 192.168.12.2) as the metric has been changed.

Practical 4C

Aim: -Implement the concept of BGP AS Path Attribute

The concept of BGP AS Path Attribute



IP and Interface Configuration of R1,R2,R3,R4

R1 >>>> IP and Network Configuration

```
R1#enable
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int f0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int loopback 0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#end
```

```
R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
*Mar 1 00:01:41.975: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:01:42.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#int f0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:02:34.827: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:02:35.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config)#int loopback 0
R1(config-if)#ip a
*Mar 1 00:03:10.543: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#end
R1#
*Mar 1 00:03:39.699: %SYS-5-CONFIG I: Configured from console by console
```

R2 >>>> IP and Network Configuration

```
R2#config t
R2(config)#int f0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 192.168.24.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int loopback 0
R2(config-if)#ip address 2.2.2.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#end
```

```
R2#enable
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:11.615: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:12.615: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#int f0/1
R2(config-if)#ip address 192.168.24.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:04:05.843: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:04:06.843: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config)#int loopback 0
R2(config-if)#ip add
*Mar 1 00:04:19.287: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)#ip address 2.2.2.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#end
R2#
*Mar 1 00:04:57.003: %SYS-5-CONFIG I: Configured from console by console
```

R3 >>>> IP and Network Configuration

```
R3#enable
R3#config t
R3(config)#int f0/1
R3(config-if)#ip address 192.168.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int f0/0
R3(config-if)#ip address 192.168.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int loopback 0
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#end
```

```
R3#enable
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int f0/1
R3(config-if)#ip address 192.168.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
*Mar 1 00:04:45.647: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:04:46.647: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config)#int f0/0
R3(config-if)#ip address 192.168.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
*Mar 1 00:06:07.139: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:08.139: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#int loopback 0
R3(config-if)#
*Mar 1 00:06:26.339: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#end
R3#
*Mar 1 00:07:08.763: %SYS-5-CONFIG_I: Configured from console by console
```

R4 >>>> IP and Network Configuration

```
R4#enable
R4#config t
R4(config)#int f0/0
R4(config-if)#ip address 192.168.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int f0/1
R4(config-if)#ip address 192.168.24.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int loopback 0
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#end
```

```
R4#enable
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)#ip address 192.168.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
*Mar 1 00:06:53.855: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:54.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#int f0/1
R4(config-if)#ip address 192.168.24.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
*Mar 1 00:07:37.239: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:07:38.239: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4(config)#int loopback 0
R4(config-if)#
*Mar 1 00:07:50.707: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#end
```

Show IP route**R1#show ip route**

```
R1#### Show IP Route
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
C    1.0.0.0/24 is subnetted, 1 subnets
C      1.1.1.0 is directly connected, Loopback0
C    192.168.13.0/24 is directly connected, FastEthernet0/1
```

R2#show ip route

```
R2#### Show IP Route
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
C    2.0.0.0/24 is subnetted, 1 subnets
C      2.2.2.0 is directly connected, Loopback0
C    192.168.24.0/24 is directly connected, FastEthernet0/1
```

R3#show ip route

```

R3#### Show IP Route
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.13.0/24 is directly connected, FastEthernet0/1
      3.0.0.0/24 is subnetted, 1 subnets
C      3.3.3.0 is directly connected, Loopback0
C    192.168.34.0/24 is directly connected, FastEthernet0/0

```

R4#show ip route

```

R4#### Show IP Route
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

      4.0.0.0/24 is subnetted, 1 subnets
C      4.4.4.0 is directly connected, Loopback0
C    192.168.24.0/24 is directly connected, FastEthernet0/1
C    192.168.34.0/24 is directly connected, FastEthernet0/0

```

Show ip int brief:**R1#show ip int brief**

```

R1#### Show IP Int Brief
R1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.12.1    YES manual up           up
Serial0/0       unassigned      YES unset   administratively down down
FastEthernet0/1 192.168.13.1    YES manual up           up
Serial0/1       unassigned      YES unset   administratively down down
Serial1/0       unassigned      YES unset   administratively down down
Serial1/1       unassigned      YES unset   administratively down down
Serial1/2       unassigned      YES unset   administratively down down
Serial1/3       unassigned      YES unset   administratively down down
Serial2/0       unassigned      YES unset   administratively down down
Serial2/1       unassigned      YES unset   administratively down down
Serial2/2       unassigned      YES unset   administratively down down
Serial2/3       unassigned      YES unset   administratively down down
Loopback0       1.1.1.1         YES manual up           up

```

R2#show ip int brief

```

R2### Show IP Int Brief
R2#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.12.2    YES manual up           up
Serial0/0       unassigned      YES unset   administratively down down
FastEthernet0/1 192.168.24.2    YES manual up           up
Serial0/1       unassigned      YES unset   administratively down down
Serial1/0       unassigned      YES unset   administratively down down
Serial1/1       unassigned      YES unset   administratively down down
Serial1/2       unassigned      YES unset   administratively down down
Serial1/3       unassigned      YES unset   administratively down down
Serial2/0       unassigned      YES unset   administratively down down
Serial2/1       unassigned      YES unset   administratively down down
Serial2/2       unassigned      YES unset   administratively down down
Serial2/3       unassigned      YES unset   administratively down down
Loopback0       2.2.2.2         YES manual up           up

```

R3#show ip int brief

```

R3#### Show IP Int Brief
R3#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.34.3    YES manual up           up
Serial0/0       unassigned      YES unset   administratively down down
FastEthernet0/1 192.168.13.3    YES manual up           up
Serial0/1       unassigned      YES unset   administratively down down
Serial1/0       unassigned      YES unset   administratively down down
Serial1/1       unassigned      YES unset   administratively down down
Serial1/2       unassigned      YES unset   administratively down down
Serial1/3       unassigned      YES unset   administratively down down
Serial2/0       unassigned      YES unset   administratively down down
Serial2/1       unassigned      YES unset   administratively down down
Serial2/2       unassigned      YES unset   administratively down down
Serial2/3       unassigned      YES unset   administratively down down
Loopback0       3.3.3.3         YES manual up           up

```

R4#show ip int brief

```
R4#### Show IP Int Brief
R4#show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.34.4	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.24.4	YES	manual	up	up
Serial0/1	unassigned	YES	unset	administratively down	down
Serial1/0	unassigned	YES	unset	administratively down	down
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down
Serial2/0	unassigned	YES	unset	administratively down	down
Serial2/1	unassigned	YES	unset	administratively down	down
Serial2/2	unassigned	YES	unset	administratively down	down
Serial2/3	unassigned	YES	unset	administratively down	down
Loopback0	4.4.4.4	YES	manual	up	up

BGP Configuration

R1#config t

R1(config)#router bgp 1

R1(config-router)#neighbor 192.168.12.2 remote-as 2

R1(config-router)#neighbor 192.168.13.3 remote-as 3

R1(config-router)#redistribute connected

R1(config-router)#end

```
R1#### BGP Config
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 1
R1(config-router)#neighbor 192.168.12.2 remote-as 2
R1(config-router)#neighbor 192.168.13.3 remote-as 3
R1(config-router)#redistribute connected
R1(config-router)#end
R1#
*Mar  1 00:24:48.991: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Mar  1 00:32:42.763: %BGP-5-ADJCHANGE: neighbor 192.168.12.2 Up
R1#
*Mar  1 00:34:55.211: %BGP-5-ADJCHANGE: neighbor 192.168.13.3 Up
```

R2#config t

R2(config)#router bgp 2

R2(config-router)#neighbor 192.168.12.1 remote-as 1

R2(config-router)#neighbor 192.168.24.4 remote-as 1

R2(config-router)#neighbor 192.168.24.4 remote-as 4

R2(config-router)#redistribute connected

R2(config-router)#end

```
R2#### BGP Config
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router bgp 2
R2(config-router)#neighbor 192.168.12.1 remote-as 1
R2(config-router)#neighbor 192.168.24.4 remote-as 1
R2(config-router)#neighbor 192.168.24.4 remote-as 4
*Mar  1 00:31:02.675: %BGP-5-ADJCHANGE: neighbor 192.168.12.1 Up
R2(config-router)#neighbor 192.168.24.4 remote-as 4
R2(config-router)#redistribute connected
R2(config-router)#end
R2#
*Mar  1 00:31:45.323: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Mar  1 00:35:08.775: %BGP-5-ADJCHANGE: neighbor 192.168.24.4 Up
```

R3#config t

R3(config)#router bgp 3

R3(config-router)#neighbor 192.168.13.1 remote-as 1

R3(config-router)#neighbor 192.168.34.4 remote-as 4

R3(config-router)#redistribute connected

R3(config-router)#end

```
R3#### BGP Config
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 3
R3(config-router)#neighbor 192.168.13.1 remote-as 1
R3(config-router)#neighbor 192.168. remote-as 1
R3(config-router)#neighbor 192.168.34.4 remote-as 4
*Mar  1 00:31:45.095: %BGP-5-ADJCHANGE: neighbor 192.168.13.1 Up
R3(config-router)#neighbor 192.168.34.4 remote-as 4
R3(config-router)#redistribute connected
R3(config-router)#end
R3#
*Mar  1 00:32:34.515: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Mar  1 00:34:27.147: %BGP-5-ADJCHANGE: neighbor 192.168.34.4 Up
```

R4#config t

R4(config)#router bgp 4

R4(config-router)#neighbor 192.168.24.2 remote-as 2

R4(config-router)#neighbor 192.168.34.3 remote-as 3

R4(config-router)#redistribute connected

R4(config-router)#end

```
R4#### BGP Config
R4#config t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router bgp 4
R4(config-router)#neighbor 192.168.24.2 remote-as 2
R4(config-router)#neighbor 192.168.34.3 remote-as 3
R4(config-router)#redistribute connected
R4(config-router)#end
R4#
*Mar  1 00:31:38.467: %BGP-5-ADJCHANGE: neighbor 192.168.24.2 Up
R4(config-router)#neighbor 192.168.34.3 remote-as 3
R4(config-router)#redistribute connected
R4(config-router)#end
R4#
*Mar  1 00:32:25.911: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 00:32:26.863: %BGP-5-ADJCHANGE: neighbor 192.168.34.3 Up
```


Creating a route map

```
R3#config t
R3(config)#route-map MED-TEST permit
R3(config-route-map)#match ip address 1
R3(config-route-map)#set metric 500
R3(config-route-map)#exit
R3(config)#route-map MED-TEST permit 20
R3(config-route-map)#end
R3#show route-map
```

```
R3#### Creating Route Map
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#route-map MED-TEST permit
R3(config-route-map)#match ip address 1
R3(config-route-map)#set metric 500
R3(config-route-map)#exit
R3(config)#route-map MED-TEST permit 20
R3(config-route-map)#end
R3#
*Mar  1 00:40:34.439: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#### Show Route Map
R3#show route-map
route-map MED-TEST, permit, sequence 10
  Match clauses:
    ip address (access-lists): 1
  Set clauses:
    metric 500
  Policy routing matches: 0 packets, 0 bytes
route-map MED-TEST, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

Creating a access list

```
R3#config t
R3(config)#access-list 1 permit host 4.4.4.4
R3(config)#end
R3#show access-list
```

```
R3### Creating Access List
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#access-list 1 permit host 4.4.4.4
R3(config)#end
R3#
*Mar  1 00:42:24.351: %SYS-5-CONFIG_I: Configured from console by console
R3#show access-list
Standard IP access list 1
  10 permit 4.4.4.4
```

Applying the created route-map to bgp configuration

```
R3#config t
R3(config)#router bgp 3
R3(config-router)#neighbor 192.168.13.1 route-map MED-TEST out
R3(config-router)#end
R3#show ip bgp
```

```
R3#### Applying created route map to BGP Config
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 3
R3(config-router)#neighbor 192.168.13.1 route-map MED-TEST out
R3(config-router)#end
R3#
*Mar  1 00:46:33.755: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#show ip bgp
BGP table version is 13, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*  1.1.1.0/24      192.168.34.4              0             0 4 2 1 ?
*>                192.168.13.1              0             0 1 ?
*  2.2.2.0/24      192.168.34.4              0             0 4 2 ?
*>                192.168.13.1              0             0 1 2 ?
*> 3.3.3.0/24      0.0.0.0                  0             32768 ?
*> 4.4.4.0/24      192.168.34.4              0             0 4 ?
*  192.168.12.0     192.168.34.4              0             0 1 2 4 ?
*>                192.168.13.1              0             0 1 ?
*> 192.168.13.0     0.0.0.0                  0             32768 ?
*  192.168.13.1     192.168.13.1              0             0 1 ?
*> 192.168.24.0     192.168.34.4              0             0 4 ?
*  192.168.13.1     192.168.13.1              0             0 1 2 ?
*  192.168.34.0     192.168.34.4              0             0 4 ?
*>                0.0.0.0                  0             32768 ?
```

R1#clear ip bgp * soft in

R1#show ip bgp

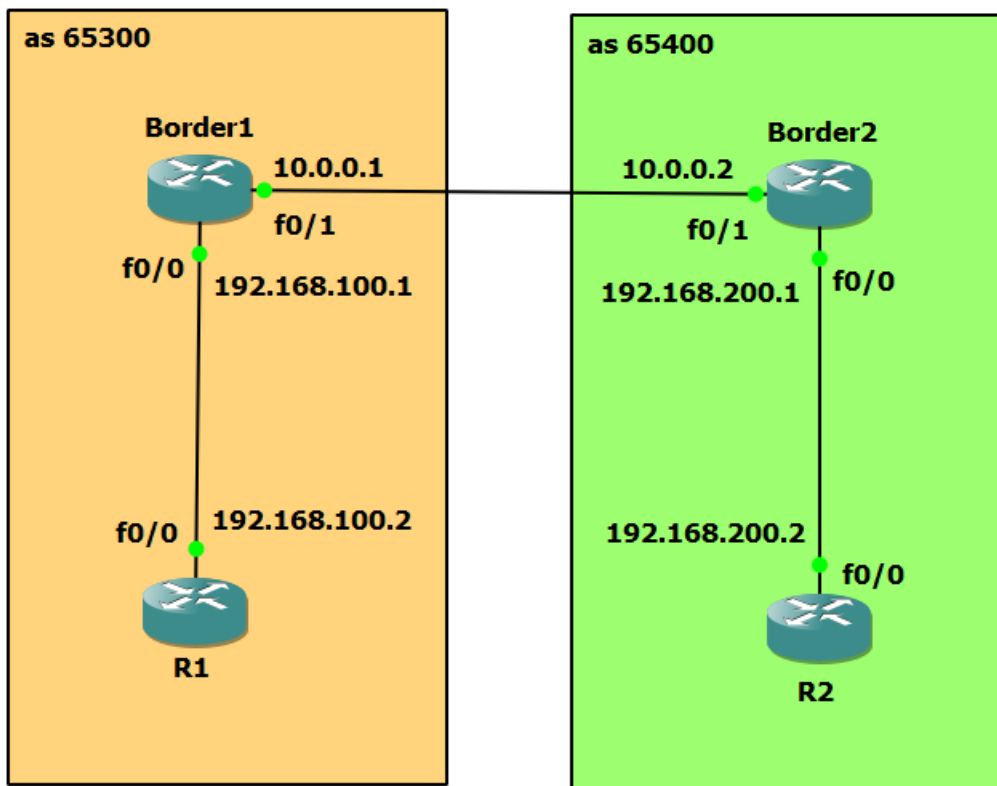
```
R1#### Clear BGP Session, Command : clear ip bgp * soft in
R1#clear ip bgp * soft in
R1#show ip bgp
BGP table version is 9, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 1.1.1.0/24      0.0.0.0                0         32768 ?
*> 2.2.2.0/24      192.168.12.2            0             0 2 ?
*> 3.3.3.0/24      192.168.13.3            0             0 3 ?
*  4.4.4.0/24      192.168.13.3            0             0 3 4 ?
*>                 192.168.12.2            0             0 2 4 ?
* 192.168.12.0     192.168.12.2            0             0 2 ?
*>                 0.0.0.0                0         32768 ?
* 192.168.13.0     192.168.13.3            0             0 3 ?
*>                 0.0.0.0                0         32768 ?
* 192.168.24.0     192.168.13.3            0             0 3 4 ?
*>                 192.168.12.2            0             0 2 ?
* 192.168.34.0     192.168.12.2            0             0 2 4 ?
*>                 192.168.13.3            0             0 3 ?
```

Practical 4 E

Aim: configure Internal BGP(IBGP) and External BGP(EBGP)

Internal BGP(IBGP) and External BGP(EBGP)



As we can see, router R1 with BORDER_1 and R2 with BORDER_2 are IBGP neighbors. BORDER_1 with BORDER_2 are EBGP neighbors and they belong to different AS.

After configure IP addresses in the interfaces, we need to configure BGP. The configuration of BGP start with router bgp AS which enters we to BGP configuration mode. The next step is configure bgp neighbors with neighbor (ip-address | peer-group-name) remote-as AS command. We will set neighbors based on ip address in this lab. Use the command network network number (mask network-mask) (route-map map-tag) to tell the router what to advertise.

Configuring IP and Interface to R1, R2, Border1, Border2

R1 >>> IP and Network Configuration

```

R1#enable
R1#config t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.100.2 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)# BGP Configuration
R1(config)#router bgp 65300
R1(config-router)#neighbor 192.168.100.1 remote-as 65300
R1(config-router)#exit
  
```

```

R1#enable
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.100.2 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:02:15.875: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:16.875: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)# BGP Configuration
R1(config)#router bgp 65300
R1(config-router)#neighbor 192.168.100.1 remote-as 65300
R1(config-router)#exit
R1(config)#
*Mar 1 00:17:18.539: %BGP-5-ADJCHANGE: neighbor 192.168.100.1 Up
  
```

Border1 >>>>

```
Border1#enable
Border1#config t
Border1(config)#int f0/1
Border1(config-if)#ip address 10.0.0.1 255.255.255.252
Border1(config-if)#no shut
Border1(config-if)#exit
Border1(config)#
Border1(config)#int f0/0
Border1(config-if)#ip address 192.168.100.1 255.255.255.0
Border1(config-if)#no shut
Border1(config-if)#exit
Border1(config)#### BGP Configuration
Border1(config)#router bgp 65300
Border1(config-router)#neighbor 192.168.100.2 remote-as 65300
Border1(config-router)#network 192.168.100.0 mask 255.255.255.0
Border1(config-router)#exit
Border1(config)#
Border1(config)#router bgp 65300
Border1(config-router)#neighbor 10.0.0.2 remote-as 65400
Border1(config-router)#exit
```

```
Border1#enable
Border1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Border1(config)#int f0/1
Border1(config-if)#ip address 10.0.0.1 255.255.255.252
Border1(config-if)#no shut
Border1(config-if)#exit
Border1(config)#
*Mar 1 00:07:59.867: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:08:00.867: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Border1(config)#int f0/0
Border1(config-if)#ip address 192.168.100.1 255.255.255.0
Border1(config-if)#no shut
Border1(config-if)#exit
Border1(config)#
*Mar 1 00:08:49.463: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:08:50.463: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Border1(config)#### BGP Configuration
Border1(config)#router bgp 65300
Border1(config-router)#neighbor 192.168.100.2 remote-as 65300
Border1(config-router)#network 192.1
*Mar 1 00:11:08.363: %BGP-5-ADJCHANGE: neighbor 192.168.100.2 Up
Border1(config-router)#network 192.168.100.0 mask 255.255.255.0
Border1(config-router)#exit
Border1(config)#
Border1(config)#router bgp 65300
Border1(config-router)#neighbor 10.0.0.2 remote-as 65400
Border1(config-router)#exit
Border1(config)#
*Mar 1 00:18:52.983: %BGP-5-ADJCHANGE: neighbor 10.0.0.2 Up
```

Border2 >>>>

```
Border2#enable
Border2#config t
Border2(config)#int f0/1
Border2(config-if)#ip address 10.0.0.2 255.255.255.252
Border2(config-if)#no shut
Border2(config-if)#exit
Border2(config)#
Border2(config)#int f0/0
Border2(config-if)#ip address 192.168.200.1 255.255.255.0
Border2(config-if)#no shut
Border2(config-if)#exit
Border2(config)#
Border2(config)### BGP Config
Border2(config)#router bgp 65400
Border2(config-router)#neighbor 192.168.100.2 remote-as 65400
Border2(config-router)#network 192.168.200.0 mask 255.255.255.0
Border2(config-router)#neighbor 10.0.0.1 remote-as 65300
Border2(config-router)#exit
```

```
Border2#enable
Border2#config t
Enter configuration commands, one per line. End with CNTL/Z.
Border2(config)#int f0/1
Border2(config-if)#ip address 10.0.0.2 255.255.255.252
Border2(config-if)#no shut
Border2(config-if)#
*Mar 1 00:10:39.535: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:10:40.535: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Border2(config-if)#exit
Border2(config)#
Border2(config)#int f0/0
Border2(config-if)#ip address 192.168.200.1 255.255.255.0
Border2(config-if)#no shut
Border2(config-if)#exit
Border2(config)#
*Mar 1 00:11:46.767: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:11:47.767: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Border2(config)### BGP Config
Border2(config)#router bgp 65400
Border2(config-router)#neighbor 192.168.100.2 remote-as 65400
Border2(config-router)#network 192.168.200.0 mask 255.255.255.0
Border2(config-router)#neighbor 10.0.0.1 remote-as 65300
Border2(config-router)#exit
Border2(config)#
*Mar 1 00:15:12.547: %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up
```

R2 >>>> IP and Network Configuration

```

R2#enable
R2#config t
R2(config)#int f0/0
R2(config-if)#ip address 192.168.200.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
R2(config) ##### BGP Config
R2(config)#router bgp 65400
R2(config-router)#neighbor 192.168.200.1 remote-as 65400
R2(config-router)#exit

```

```

R2#enable
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.200.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:13:13.259: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:13:14.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#
R2(config) ##### BGP Config
R2(config)#router bgp 65400
R2(config-router)#neighbor 192.168.200.1 remote-as 65400
R2(config-router)#exit

```

If we use the mask in network command then the network in routing table must match network from this command. Also note that we can advertise any network that resides in routing table and is not necessary to be directly connected.

Verify BGP Status:

R1(config)#do show ip bgp

```

R1(config) ##### Show IP BGP
R1(config)#do show ip bgp
BGP table version is 3, local router ID is 192.168.100.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
r>i192.168.100.0    192.168.100.1           0      100      0 i
* i192.168.200.0    10.0.0.2                0      100      0 65400 i

```

R1(config)#do show ip bgp summary

```

R1(config)#do show ip bgp summary
BGP router identifier 192.168.100.2, local AS number 65300
BGP table version is 3, main routing table version 3
2 network entries using 240 bytes of memory
2 path entries using 104 bytes of memory
3/1 BGP path/bestpath attribute entries using 372 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 740 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.100.1  4 65300    22     20      3     0   0 00:16:08      2

```

Border1(config)#do show ip bgp

```

Border1(config)#do show ip bgp
BGP table version is 3, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
*> 192.168.100.0    0.0.0.0              0         32768 i
*> 192.168.200.0    10.0.0.2             0           0 65400 i

```

Border1(config)#do show ip bgp summary

```

Border1(config)#do show ip bgp summary
BGP router identifier 192.168.100.1, local AS number 65300
BGP table version is 3, main routing table version 3
2 network entries using 240 bytes of memory
2 path entries using 104 bytes of memory
3/2 BGP path/bestpath attribute entries using 372 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 2 (at peak 2) using 64 bytes of memory
BGP using 804 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.2       4 65400    13     13      3     0   0 00:08:54      1
192.168.100.2  4 65300    20     22      3     0   0 00:16:39      0

```

You can see that we have two neighbors with BORDER_1, one internal and one external. Let's see the results on R1:

R1(config)#do show ip route

```
R1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.100.0/24 is directly connected, FastEthernet0/0
```

As we can see on R1 we don't have external route to network 192.168.100.0/24. This is because BORDER_2, which is an external router, routing updates sent have address 10.0.0.2 as source ip address, but in this case R1 doesn't have any idea how to get 10.0.0.0 network. BGP on router R1 doesn't put in routing table a route to 192.168.100.0/24. To overcome this we have to add on router BORDER_1 this command: neighbor 192.168.0.2 next-hop-self, this command will change source ip address of routing updates sent to router R1 (will change to its e0/1 ip address). We need to put the command inside router bgp AS configuration.

Border1(config)#### Applying Next hop self to BGP

Border1(config)#router bgp 65300

Border1(config-router)#neighbor 192.168.100.2 remote-as 65300

Border1(config-router)#neighbor 192.168.100.2 next-hop-self

Border1(config-router)#exit

```
Border1(config)#### Applying Next hop self to BGP
Border1(config)#router bgp 65300
Border1(config-router)#neighbor 192.168.100.2 remote-as 65300
Border1(config-router)#neighbor 192.168.100.2 next-hop-self
Border1(config-router)#exit
```

We need to do the same on router BORDER_2 to change update source to neighbor R1.

Border2(config)#### Applying next hop self to BGP

Border2(config)#router bgp 65400

Border2(config-router)#neighbor 192.168.200.2 remote-as 65400

Border2(config-router)#neighbor 192.168.200.2 next-hop-self

Border2(config-router)#exit

```
Border2(config)#### Applying next hop self to BGP
Border2(config)#router bgp 65400
Border2(config-router)#neighbor 192.168.200.2 remote-as 65400
Border2(config-router)#neighbor 192.168.200.2 next-hop-self
*Mar  1 00:41:58.979: %BGP-5-ADJCHANGE: neighbor 192.168.200.2 Up
Border2(config-router)#neighbor 192.168.200.2 next-hop-self
Border2(config-router)#exit
Border2(config)#
Border2#
*Mar  1 00:53:52.963: %SYS-5-CONFIG_I: Configured from console by console
```

So, let's see again routing table on router R1

R1(config)#### After applying next-hop-self to BGP

R1(config)#do show ip route

```
R1(config)#### After applying next-hop-self to BGP
R1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

B    192.168.200.0/24 [200/0] via 192.168.100.1, 00:03:11
C    192.168.100.0/24 is directly connected, FastEthernet0/0
```


Then, ping from R1 to R2

R1#ping 192.168.200.2

```
R1#ping 192.168.200.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.200.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/88/96 ms
```

Ping From Border1 to Border2

Border1#ping 192.168.200.1

```
Border1#ping 192.168.200.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.200.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/28/32 ms
```

Ping from R2 to R1

R2#ping 192.168.100.2

```
R2#ping 192.168.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 84/90/92 ms
```

Ping from R2 to Border1

R2#ping 192.168.100.1

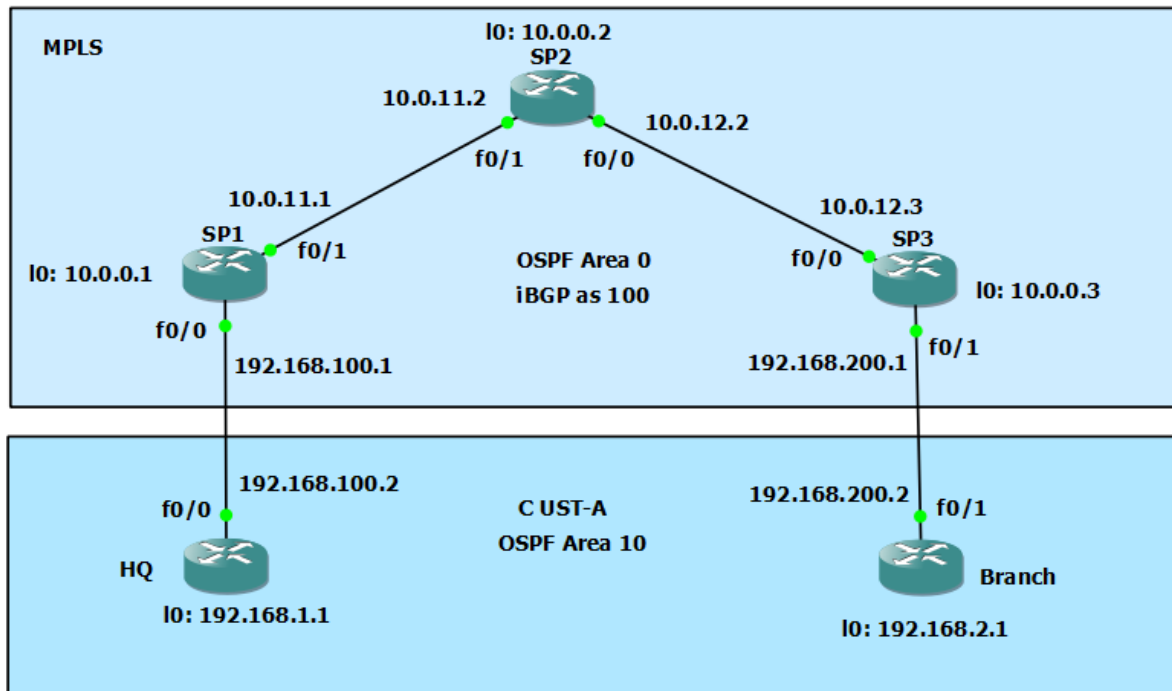
```
R2#ping 192.168.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/57/72 ms
```

Practical 6A

Aim: Implementing MPLS VPNs

Implementing MPLS VPNs



Configure IP and Interfaces of SP1, SP2, SP3, HQ and Branch Routers.

```
SP1#config t
SP1(config)#int loopback 0
SP1(config-if)#ip address 10.0.0.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#int f0/1
SP1(config-if)#ip address 10.0.11.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#int f0/0
SP1(config-if)#ip address 192.168.100.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#exit
```

```
SP1#enable
SP1#config t
Enter configuration commands, one per line. End with CNTL/Z.
SP1(config)#int loopback 0
SP1(config-if)#ip a
*Mar 1 00:00:49.139: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
SP1(config-if)#ip address 10.0.0.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#int f0/1
SP1(config-if)#ip address 10.0.11.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#int
*Mar 1 00:01:56.147: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:01:57.147: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
SP1(config-if)#int f0/0
SP1(config-if)#192.168.100.1 255.255.255.0
^
% Invalid input detected at '^' marker.

SP1(config-if)#ip address 192.168.100.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#
*Mar 1 00:03:09.043: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:10.043: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
SP1(config-if)#exit
```

```
SP2#enable
SP2#config t
SP2(config)#int loopback 0
SP2(config-if)#ip address 10.0.0.2 255.255.255.0
SP2(config-if)#no shut
SP2(config-if)#int f0/1
SP2(config-if)#ip address 10.0.11.2 255.255.255.0
SP2(config-if)#no shut
SP2(config-if)#int f0/0
SP2(config-if)#ip address 10.0.12.2 255.255.255.0
SP2(config-if)#no shut
SP2(config-if)#exit
```

```
SP2#enable
SP2#config t
Enter configuration commands, one per line. End with CNTL/Z.
SP2(config)#int loopback 0
SP2(config-if)#ip
*Mar 1 00:02:12.479: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
SP2(config-if)#ip address 10.0.0.2 255.255.255.0
SP2(config-if)#no shut
SP2(config-if)#int f0/1
SP2(config-if)#ip address 10.0.11.2 255.255.255.0
SP2(config-if)#no shut
SP2(config-if)#int f
*Mar 1 00:03:11.803: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:12.803: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state
to up
SP2(config-if)#int f0/0
SP2(config-if)#ip address 10.0.12.2 255.255.255.0
SP2(config-if)#no shut
SP2(config-if)#
*Mar 1 00:04:00.907: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:04:01.907: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to up
SP2(config-if)#exit
```

```
SP3#enable
SP3#config t
SP3(config)#int loopback 0
SP3(config-if)#ip address 10.0.0.3 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#int f0/0
SP3(config-if)#ip address 10.0.12.3 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#int f0/1
SP3(config-if)#ip address 192.168.200.1 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#exit
```

```
SP3#enable
SP3#config t
Enter configuration commands, one per line. End with CNTL/Z.
SP3(config)#int loopback 0
SP3(config-if)#ip ad
*Mar 1 00:03:26.499: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
SP3(config-if)#ip address 10.0.0.3 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#int f0/0
SP3(config-if)#ip address 10.0.12.3 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#int
*Mar 1 00:04:27.859: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:04:28.859: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to up
SP3(config-if)#int f0/1
SP3(config-if)#ip address 192.168.200.1 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#
*Mar 1 00:05:13.727: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:05:14.727: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state
to up
SP3(config-if)#exit
```

```
HQ#enable
HQ#config t
HQ(config)#int loopback 0
HQ(config-if)#ip address 192.168.1.1 255.255.255.0
HQ(config-if)#no shut
HQ(config-if)#int f0/0
HQ(config-if)#ip address 192.168.100.2 255.255.255.0
HQ(config-if)#no shut
HQ(config-if)#exit
```

```
HQ#enable
HQ#config t
Enter configuration commands, one per line. End with CNTL/Z.
HQ(config)#int loopback 0
HQ(config-if)#ip a
*Mar 1 00:04:40.187: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
HQ(config-if)#ip address 192.168.1.1 255.255.255.0
HQ(config-if)#no shut
HQ(config-if)#int f0/0
HQ(config-if)#ip address 192.168.100.2 255.255.255.0
HQ(config-if)#no shut
HQ(config-if)#
*Mar 1 00:06:00.015: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:01.015: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to up
HQ(config-if)#exit
```

```
Branch#enable
Branch#config t
Branch(config)#int loopback 0
Branch(config-if)#ip address 192.168.2.1 255.255.255.0
Branch(config-if)#no shut
Branch(config-if)#int f0/1
Branch(config-if)#ip address 192.168.200.2 255.255.255.0
Branch(config-if)#no shut
Branch(config-if)#exit
```

```
Branch#enable
Branch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Branch(config)#int loopback 0
Branch(config-if)#
*Mar 1 00:06:05.015: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
Branch(config-if)#ip address 192.168.2.1 255.255.255.0
Branch(config-if)#no shut
Branch(config-if)#int f0/1
Branch(config-if)#ip address 192.168.200.2 255.255.255.0
Branch(config-if)#no shut
Branch(config-if)#
*Mar 1 00:07:03.023: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:07:04.023: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state
to up
Branch(config-if)#exit
```

Ospf Configuration

```
SP1(config)#router ospf 1
SP1(config-router)#network 10.0.0.0 0.0.0.255 area 0
SP1(config-router)#network 10.0.11.0 0.0.0.255 area 0
SP1(config-router)#exit
SP1(config)#do show ip ospf neighbor
```

```
SP1(config)#### OSPF Configuration
SP1(config)#router ospf 1
SP1(config-router)#network 10.0.0.0 0.0.0.255 area 0
SP1(config-router)#network 10.0.11.0 0.0.0.255 area 0
SP1(config-router)#exit
SP1(config)#
SP1(config)#
*Mar 1 00:23:20.599: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.2 on FastEthernet0/1 from LOADING to FULL, Loading Done
SP1(config)#### Checking OSPF Neighbor
SP1(config)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.2	1	FULL/BDR	00:00:36	10.0.11.2	FastEthernet0/1

```
SP2(config)#router ospf 1
SP2(config-router)#network 10.0.0.0 0.0.0.255 area 0
SP2(config-router)#network 10.0.11.0 0.0.0.255 area 0
SP2(config-router)#network 10.0.12.0 0.0.0.255 area 0
SP2(config-router)#exit
SP2(config)#do show ip ospf neighbor
```

```
SP2(config)### OSPF Config
SP2(config)#router ospf 1
SP2(config-router)#network 10.0.0.0 0.0.0.255 area 0
SP2(config-router)#network 10.0.11.0 0.0.0.255 area 0
SP2(config-router)#network 10.0.12.0 0.0.0.255 area 0
*Mar 1 00:22:00.547: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.1 on FastEthernet0/1 from LOADING to FULL, Loading Done
SP2(config-router)#network 10.0.12.0 0.0.0.255 area 0
SP2(config-router)#exit
SP2(config)#
*Mar 1 00:24:14.991: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.3 on FastEthernet0/0 from LOADING to FULL, Loading Done
SP2(config)#### Checking OSPF Neighbor
SP2(config)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.3	1	FULL/BDR	00:00:34	10.0.12.3	FastEthernet0/0
10.0.0.1	1	FULL/DR	00:00:39	10.0.11.1	FastEthernet0/1

```
SP3(config)#router ospf 1
SP3(config-router)#network 10.0.0.0 0.0.0.255 area 0
SP3(config-router)#network 10.0.12.0 0.0.0.255 area 0
SP3(config-router)#exit
SP3(config)#do show ip ospf neighbor
```

```
SP3(config)### OSPF config
SP3(config)#router ospf 1
SP3(config-router)#network 10.0.0.0 0.0.0.255 area 0
SP3(config-router)#network 10.0.12.0 0.0.0.255 area 0
SP3(config-router)#exit
SP3(config)#
*Mar 1 00:22:14.767: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
SP3(config)#### Checkinng OSPF neighbor
SP3(config)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.2	1	FULL/DR	00:00:33	10.0.12.2	FastEthernet0/0

MPLS config

```
SP1(config)#mpls ldp router-id loopback 0 force
SP1(config)#int f0/1
SP1(config-if)#mpls ip
SP1(config-if)#no shut
SP1(config-if)#exit
SP1(config)#do show mpls ldp neighbor
```

```
SP1(config)#### MPLS Configurarion
SP1(config)#mpls ldp router-id loopback 0 force
SP1(config)#int f0/1
SP1(config-if)#mpls ip
SP1(config-if)#no shut
SP1(config-if)#exit
SP1(config)#
*Mar 1 00:42:28.767: %LDP-5-NBRCHG: LDP Neighbor 10.0.0.2:0 (1) is UP
SP1(config)#### Show MPLS LDP Neighbor
SP1(config)#do show mpls ldp neighbor
  Peer LDP Ident: 10.0.0.2:0; Local LDP Ident 10.0.0.1:0
    TCP connection: 10.0.0.2.33797 - 10.0.0.1.646
    State: Oper; Msgs sent/rcvd: 32/31; Downstream
    Up time: 00:19:59
    LDP discovery sources:
      FastEthernet0/1, Src IP addr: 10.0.11.2
    Addresses bound to peer LDP Ident:
      10.0.12.2      10.0.11.2      10.0.0.2
```

```

SP2(config)#mpls ldp router-id loopback 0 force
SP2(config)#int f0/1
SP2(config-if)#mpls ip
SP2(config-if)#no shut
SP2(config-if)#exit
SP2(config)#mpls ldp router-id loopback 0 force
SP2(config)#int f0/0
SP2(config-if)#mpls ip
SP2(config-if)#no shut
SP2(config-if)#exit
SP2(config)#do show mpls ldp neighbor

```

```

SP2(config)#### MPLS Configuration
SP2(config)#mpls ldp router-id loopback 0 force
SP2(config)#int f0/1
SP2(config-if)#mpls ip
SP2(config-if)#no shu
*Mar 1 00:41:08.647: %LDP-5-NBRCHG: LDP Neighbor 10.0.0.1:0 (1) is UP
SP2(config-if)#no shut
SP2(config-if)#exit
SP2(config)#mpls ldp router-id loopback 0 force
SP2(config)#int f0/0
SP2(config-if)#mpls ip
SP2(config-if)#no shut
SP2(config-if)#exit

```

```

SP2(config)#### Show mpls ldp Neighbor
SP2(config)#do show mpls ldp neighbor
  Peer LDP Ident: 10.0.0.1:0; Local LDP Ident 10.0.0.2:0
    TCP connection: 10.0.0.1.646 - 10.0.0.2.33797
    State: Oper; Msgs sent/rcvd: 15/16; Downstream
    Up time: 00:06:09
    LDP discovery sources:
      FastEthernet0/1, Src IP addr: 10.0.11.1
    Addresses bound to peer LDP Ident:
      192.168.100.1  10.0.11.1  10.0.0.1
  Peer LDP Ident: 10.0.0.3:0; Local LDP Ident 10.0.0.2:0
    TCP connection: 10.0.0.3.16355 - 10.0.0.2.646
    State: Oper; Msgs sent/rcvd: 11/12; Downstream
    Up time: 00:03:05
    LDP discovery sources:
      FastEthernet0/0, Src IP addr: 10.0.12.3
    Addresses bound to peer LDP Ident:
      10.0.12.3  192.168.200.1  10.0.0.3

```

```

SP3(config)#mpls ldp router-id loopback 0 force
SP3(config)#int f0/0
SP3(config-if)#mpls ip
SP3(config-if)#no shut
SP3(config)#do show mpls ldp neighbor

```

```

SP3(config)#### MPLS Config
SP3(config)#mpls ldp router-id loopback 0 force
SP3(config)#int f0/0
SP3(config-if)#mpls ip
SP3(config-if)#no shut
SP3(config-if)#e
*Mar 1 00:42:12.071: %LDP-5-NBRCHG: LDP Neighbor 10.0.0.2:0 (1) is UP
SP3(config-if)#exit
SP3(config)### Show MPLS LDP Neighbor
SP3(config)#do show mpls ldp neighbor
  Peer LDP Ident: 10.0.0.2:0; Local LDP Ident 10.0.0.3:0
    TCP connection: 10.0.0.2.646 - 10.0.0.3.16355
    State: Oper; Msgs sent/rcvd: 28/28; Downstream
    Up time: 00:17:55
    LDP discovery sources:
      FastEthernet0/0, Src IP addr: 10.0.12.2
    Addresses bound to peer LDP Ident:
      10.0.12.2  10.0.11.2  10.0.0.2

```


Config the vrf CUST-A

```
SP1(config)#ip vrf CUST-A
SP1(config-vrf)#rd 100:1
SP1(config-vrf)#route-target export 1:100
SP1(config-vrf)#route-target import 1:100
SP1(config-vrf)#!
SP1(config-vrf)#exit
SP1(config)#int f0/0
SP1(config-if)#ip vrf forwarding CUST-A
SP1(config-if)#ip address 192.168.100.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#exit
```

```
SP1(config)### Configuring VRF (Virtual Routing and Forwarding) for CUST-A
SP1(config)#ip vrf CUST-A
SP1(config-vrf)#rd 100:1
SP1(config-vrf)#route-target export 1:100
SP1(config-vrf)#route-target import 1:100
SP1(config-vrf)#!
SP1(config-vrf)#exit
```

```
SP1(config)#int f0/0
SP1(config-if)#ip vrf forwarding CUST-A
% Interface FastEthernet0/0 IP address 192.168.100.1 removed due to enabling VRF CUST-A
SP1(config-if)#ip address 192.168.100.1 255.255.255.0
SP1(config-if)#no shut
SP1(config-if)#exit
```

```
SP3(config)#ip vrf CUST-A
SP3(config-vrf)#rd 100:1
SP3(config-vrf)#route-target export 1:100
SP3(config-vrf)#route-target import 1:100
SP3(config-vrf)#!
SP3(config-vrf)#exit
SP3(config)#int f0/1
SP3(config-if)#ip vrf forwarding CUST-A
SP3(config-if)#ip address 192.168.200.1 255.255.255.0
SP3(config-if)#no shut
SP3(config-if)#exit
```

Checking the vrf

```
SP1(config)#do ping vrf CUST-A 192.168.100.2
SP1(config)#### Checking the VRF, ping 192.168.100.2(HQ) from SP1
SP1(config)#do ping vrf CUST-A 192.168.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 28/32/36 ms
```

```
SP3(config)#do ping vrf CUST-A 192.168.200.2
SP3(config)### Checking the VRF, ping 192.168.200.2(Branch)
SP3(config)#do ping vrf CUST-A 192.168.200.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.200.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 32/41/60 ms
```

Customer routing

```
HQ(config)#router ospf 1
HQ(config-router)#log-adjacency-changes
HQ(config-router)#network 192.168.1.0 0.0.0.255 area 10
HQ(config-router)#network 192.168.100.0 0.0.0.255 area 10
HQ(config-router)#exit
```

```
HQ(config)### OSPF Configurationn
HQ(config)#router ospf 1
HQ(config-router)#log-adjacency-changes
HQ(config-router)#network 192.168.1.0 0.0.0.255 area 10
HQ(config-router)#network 192.168.100.0 0.0.0.255 area 10
HQ(config-router)#exit
HQ(config)#
*Mar 1 01:37:18.175: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.1 on FastEthernet0/0 from LOADING t
o FULL, Loading Done
```

```
Branch(config)#router ospf 1
Branch(config-router)#log-adjacency-changes
Branch(config-router)#network 192.168.2.0 0.0.0.255 area 10
Branch(config-router)#network 192.168.200.0 0.0.0.255 area 10
Branch(config-router)#exit
```

```
Branch(config)### OSPF Configuration
Branch(config)#router ospf 1
Branch(config-router)#log-adjacency-changes
Branch(config-router)#network 192.168.2.0 0.0.0.255 area 10
Branch(config-router)#network 192.168.200.0 0.0.0.255 area 10
Branch(config-router)#exit
Branch(config)#
*Mar 1 01:39:07.163: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.200.1 on FastEthernet0/1 from LOADING t
o FULL, Loading Done
```

```
SP1(config)#### OSPF Configurationn to CUST A
SP1(config)#router ospf 10 vrf CUST-A
SP1(config-router)#network 192.168.100.0 0.0.0.255 area 10
SP1(config-router)#exit
SP1(config)#do show ip ospf neighbor
```

```
SP1(config)### OSPF Configurationn to CUST A
SP1(config)#router ospf 10 vrf CUST-A
SP1(config-router)#network 192.168.100.0 0.0.0.255 area 10
SP1(config-router)#
*Mar 1 01:43:18.411: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.1.1 on FastEthernet0/0 from LOADING to
FULL, Loading Done
```

```
SP1(config)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.0.2	1	FULL/BDR	00:00:32	10.0.11.2	FastEthernet0/1
192.168.1.1	1	FULL/DR	00:00:35	192.168.100.2	FastEthernet0/0

```
SP1(config)#do show ip route vrf CUST-A
```

```
SP1(config)#### Show IP Route VRF CUST A
SP1(config)#do show ip route vrf CUST-A

Routing Table: CUST-A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/32 is subnetted, 1 subnets
O       192.168.1.1 [110/11] via 192.168.100.2, 00:07:17, FastEthernet0/0
C       192.168.100.0/24 is directly connected, FastEthernet0/0
```

```
SP3(config)#router ospf 10 vrf CUST-A
```

```
SP3(config-router)#network 192.168.200.0 0.0.0.255 area 10
```

```
SP3(config-router)#exit
```

```
SP3(config)### OSPF Configuration to CUST A
SP3(config)#router ospf 10 vrf CUST-A
SP3(config-router)#network 192.168.200.0 0.0.0.255 area 10
SP3(config-router)#
*Mar  1 01:42:47.387: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on FastEthernet0/1 from LOADING to
FULL, Loading Done
```

```
SP3(config)#do show ip ospf neighbor
```

```
SP3(config)### OSPF Configuration to CUST A
SP3(config)#router ospf 10 vrf CUST-A
SP3(config-router)#network 192.168.200.0 0.0.0.255 area 10
SP3(config-router)#
*Mar  1 01:42:47.387: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on FastEthernet0/1 from LOADING to
FULL, Loading Done
```

```
SP3(config)#do show ip route vrf CUST-A
```

```
SP3(config)#do show ip route vrf CUST-A

Routing Table: CUST-A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.200.0/24 is directly connected, FastEthernet0/1
     192.168.2.0/32 is subnetted, 1 subnets
O      192.168.2.1 [110/11] via 192.168.200.2, 00:13:16, FastEthernet0/1
```

BGP

```
SP1(config)#router bgp 100
```

```
SP1(config-router)#neighbor 10.0.0.3 remote-as 100
```

```
SP1(config-router)#neighbor 10.0.0.3 update-source loopback 0
```

```
SP1(config-router)#exit
```

```
SP1(config)#do show ip bgp summary
```

```
SP1(config)### BGP Configuration
SP1(config)#router bgp 100
SP1(config-router)#neighbor 10.0.0.3 remote-as 100
SP1(config-router)#neighbor 10.0.0.3 update-source loopback 0
SP1(config-router)#exit
SP1(config)#do show ip bgp summary
BGP router identifier 10.0.0.1, local AS number 100
BGP table version is 1, main routing table version 1

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.3      4   100      8      8        1    0    0 00:04:43      0
SP1(config)#
```

```

SP3(config)#router bgp 100
SP3(config-router)#neighbor 10.0.0.1 remote-as 100
SP3(config-router)#neighbor 10.0.0.1 update-source loopback 0
SP3(config-router)#exit
SP3(config)#do show ip bgp summary

```

```

SP3(config)#### BGP Configuration
SP3(config)#router bgp 100
SP3(config-router)#neighbor 10.0.0.1 remote-as 100
SP3(config-router)#neighbor 10.0.0.1
*Mar 1 02:12:37.583: %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up
SP3(config-router)#neighbor 10.0.0.1 update-source loopback 0
SP3(config-router)#exit
SP3(config)#do show ip bgp summary
BGP router identifier 10.0.0.3, local AS number 100
BGP table version is 1, main routing table version 1

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.1      4    100      4      4        1    0   0 00:00:58      0
SP3(config)#
*Mar 1 02:23:34.791: %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Down Peer closed the session
SP3(config)#
*Mar 1 02:23:36.035: %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up

```

```

SP1(config)#router bgp 100
SP1(config-router)#address-family vpnv4
SP1(config-router-af)#neighbor 10.0.0.3 activate

```

```

SP1(config)### address family VPNv4 Configuration
SP1(config)#router bgp 100
SP1(config-router)#address-family vpnv4
SP1(config-router-af)#neighbor 10.0.0.3 activate
SP1(config-router-af)#
*Mar 1 02:26:54.075: %BGP-5-ADJCHANGE: neighbor 10.0.0.3 Down Address family activated
SP1(config-router-af)#
*Mar 1 02:26:56.259: %BGP-5-ADJCHANGE: neighbor 10.0.0.3 Up

```

```

SP3(config)#router bgp 100
SP3(config-router)#address-family vpnv4
SP3(config-router-af)#neighbor 10.0.0.1 activate

```

```

SP3(config)### Address Family VPNv4 configuration
SP3(config)#router bgp 100
SP3(config-router)#address-family vpnv4
SP3(config-router-af)#neighbor 10.0.0.1 activate
SP3(config-router-af)#
*Mar 1 02:26:02.799: %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Down Address family activated
SP3(config-router-af)#
*Mar 1 02:26:04.987: %BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up

```

```

SP1(config-router-af)#### Applying IPv4 to CUST-A
SP1(config-router-af)#router ospf 10 vrf CUST-A
SP1(config-router)#redistribute bgp 100 subnets
SP1(config-router)#exit
SP1(config)#
SP1(config)#router bgp 100
SP1(config-router)#address-family ipv4 vrf CUST-A
SP1(config-router-af)#redistribute ospf 10 vrf CUST-A
SP1(config-router-af)#exit-address-family
SP1(config-router)#exit

```

```

SP1(config-router-af)#### Applying IPv4 to CUST-A
SP1(config-router-af)#router ospf 10 vrf CUST-A
SP1(config-router)#redistribute bgp 100 subnets
SP1(config-router)#exit
SP1(config)#
SP1(config)#router bgp 100
SP1(config-router)#address-family ipv4 vrf CUST-A
SP1(config-router-af)#redistribute ospf 10 vrf CUST-A
SP1(config-router-af)#
*Mar 1 02:40:46.271: %BGP-4-VPNv4NH_MASK: Nexthop 10.0.0.1 may not be reachable from neighbor 10.0.0.3 - not /32 mask
SP1(config-router-af)#exit-address-family
SP1(config-router)#exit

```

```

SP3(config)### Applying IPv4 tp CUST A
SP3(config)#router ospf 10 vrf CUST-A
SP3(config-router)#redistribute bgp 100 subnets
SP3(config-router)#exit
SP3(config)#
SP3(config)#router bgp 100
SP3(config-router)#address-family ipv4 vrf CUST-A
SP3(config-router-af)#redistribute ospf 10 vrf CUST-A
SP3(config-router-af)#exit-address-family
SP3(config-router)#exit

```

```

SP3(config)### Applying IPv4 tp CUST A
SP3(config)#router ospf 10 vrf CUST-A
SP3(config-router)#redistribute bgp 100 subnets
SP3(config-router)#exit
SP3(config)#
SP3(config)#router bgp 100
SP3(config-router)#address-family ipv4 vrf CUST-A
SP3(config-router-af)#redistribute ospf 10 vrf CUST-A
SP3(config-router-af)#
*Mar 1 02:48:18.871: %BGP-4-VPNv4NH_MASK: Nexthop 10.0.0.3 may not be reachable from neighbor 10.0.0.1 - not /32 mask
SP3(config-router-af)#exit-address-family
SP3(config-router)#exit

```

```

SP1(config)### Show IP BGP VPNv4 all
SP1(config)#do show ip bgp vpnv4 all

```

```

SP1(config)### Show IP BGP VPNv4 all
SP1(config)#do show ip bgp vpnv4 all
BGP table version is 9, local router ID is 10.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf CUST-A)
*> 192.168.1.1/32    192.168.100.2           11             32768 ?
*>i192.168.2.1/32    10.0.0.3                11            100      0 ?
*> 192.168.100.0     0.0.0.0                 0             32768 ?
*>i192.168.200.0     10.0.0.3                0            100      0 ?
SP1(config)#

```

```

SP1(config)### Show ip route vrf CUST A
SP1(config)#do show ip route vrf CUST-A

```

```

SP1(config)### Show ip route vrf CUST A
SP1(config)#do show ip route vrf CUST-A

Routing Table: CUST-A
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

B    192.168.200.0/24 [200/0] via 10.0.0.3, 00:04:27
     192.168.1.0/32 is subnetted, 1 subnets
O     192.168.1.1 [110/11] via 192.168.100.2, 01:12:50, FastEthernet0/0
     192.168.2.0/32 is subnetted, 1 subnets
B     192.168.2.1 [200/11] via 10.0.0.3, 00:04:27
C    192.168.100.0/24 is directly connected, FastEthernet0/0
SP1(config)#

```

SP1(config)#do show ip route

```
SP1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.11.0/24 is directly connected, FastEthernet0/1
O       10.0.12.0/24 [110/20] via 10.0.11.2, 02:33:24, FastEthernet0/1
O       10.0.0.2/32 [110/11] via 10.0.11.2, 02:33:34, FastEthernet0/1
O       10.0.0.3/32 [110/21] via 10.0.11.2, 02:31:10, FastEthernet0/1
C       10.0.0.0/24 is directly connected, Loopback0
```

SP2(config)#do show ip route

```
SP2(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.11.0/24 is directly connected, FastEthernet0/1
C       10.0.12.0/24 is directly connected, FastEthernet0/0
O       10.0.0.3/32 [110/11] via 10.0.12.3, 02:32:49, FastEthernet0/0
C       10.0.0.0/24 is directly connected, Loopback0
O       10.0.0.1/32 [110/11] via 10.0.11.1, 02:35:00, FastEthernet0/1
SP2(config)#
```

SP3(config)#do show ip route

```
SP3(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O       10.0.11.0/24 [110/20] via 10.0.12.2, 02:32:36, FastEthernet0/0
C       10.0.12.0/24 is directly connected, FastEthernet0/0
O       10.0.0.2/32 [110/11] via 10.0.12.2, 02:32:36, FastEthernet0/0
C       10.0.0.0/24 is directly connected, Loopback0
O       10.0.0.1/32 [110/21] via 10.0.12.2, 02:32:36, FastEthernet0/0
SP3(config)#
```